

Supplementary Information (Online Resource 1)

for:

Auditing and Partially Reproducing the SLRTP2025 Back-Translation Evaluator

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Overview

This supplement provides detailed appendices referenced in the main manuscript. Section labels match the labels used in the main text (e.g., `sec:appendix missing_id` → Sup. Section 1).

1 Missing Test ID Sensitivity

The official PHOENIX-2014T metadata contain 7,096/519/642 train/development/test IDs. The released SLRTP2025 pose records available to this study contain 7,060/515/641 sequences: 36/4/1 official IDs are absent from the released pose materialization. Their IDs are enumerated in `artifact/manifests/exclusions.jsonl`, and the ID-level reconciliation of `results/split_reconciliation.json` (SI Sup. V) classifies them: all 41 are present in an independent upstream skeleton extraction carrying the full official counts (7,096/519/642), so the drop occurred inside the SLRTP2025 pose packaging rather than upstream; the dropped items show no duration or signer anomaly (train frames 22–262 across six of nine signers), and only two of the 36 dropped training texts lack an exact copy in the retained pool (both high-frequency template sentences). The shipped `train.pt` contains exactly 7,060 items, so the released training manifest is itself the reduced pool. The missing test ID is `13December_2010_Monday_tagesschau-6940`; its text/gloss metadata are not present in the released pose record, so length and template density cannot be reconstructed without an additional official annotation source. We therefore condition every result on the complete 641-record release rather than assert that the missing item is ignorable. As an empirical observed-extreme sensitivity

scenario, assigning the missing item the per-item score of the largest PURE-over-REC outlier observed among the 641 released items changes the corpus-level gap by less than ± 0.2 BLEU points, well within the reported 95% paired-bootstrap interval $[+9.6, +12.4]$. This is an observed-extreme scenario, not a worst-case bound: corpus BLEU is non-linear and the missing item’s reference, hypothesis, and n-gram sufficient statistics are unknown, so the true mathematical worst case is not covered. All primary conclusions are therefore restricted to the released 641-record materialization.

2 Per-seed Legacy Cells (101–606 primary; 707–1405 extension)

The six primary seeds (101–606) were prospectively specified before their results were inspected; the eight extension seeds are post-hoc additions that enter only the within-implementation seed dispersion interval and descriptive ranges, never the six primary comparisons. Per-seed PHX-public values (sacreBLEU 0–100 scale, canonical §3.2 donor registry). PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1; gaps are relative to local REC:

Table S1: Prospectively specified legacy seeds (101–606): canonical fixed-output evaluation on the 641-pose PHX-public panel.

Evaluator	REC	PURE	Gap	COMP	Gap
Released	12.78	23.02	+10.24	18.86	+6.09
Seed 101	9.24	8.45	−0.80	7.92	−0.65
Seed 202	8.22	6.92	−1.30	7.54	−1.28
Seed 303	9.10	7.25	−1.85	8.71	−0.96
Seed 404	9.52	7.71	−1.81	6.38	−0.81
Seed 505	9.71	8.44	−1.28	7.97	−1.08
Seed 606	7.55	6.31	−1.24	7.93	−0.16

Seed-stratified dispersion intervals and two-level resampling.

The 14-seed within-implementation seed dispersion interval splits into the six internally prespecified primary seeds and the eight post-hoc extension seeds: the primary-six give a Student- t interval of $[-2.48, -0.28]$ (mean -1.38 , SD 0.40 , 0/6 positive) and the extension-eight $[-1.61, -0.54]$ (mean -1.07 , SD 0.21 , 0/8 positive). Leave-one-seed-out refits keep the interval below zero in all 14 folds (lower bounds -1.99 to -1.80 ; upper bounds -0.51 to -0.42). A two-level bootstrap resamples seeds with replacement (14 draws), then resamples each drawn seed’s 641 items with replacement and averages the corpus gap over the 14 drawn seeds, giving a 95% CI of $[-1.49, -0.93]$ (10,000 resamples, seed 42; per-item BLEU sufficient statistics precomputed once per seed, so the resamples are numerically identical to recomputing corpus BLEU per draw). Every seed-level resampling leaves the non-replication conclusion unchanged.

Table S2: Per-seed extension cells for the eight post-hoc extension seeds (707–1405). All values are PHX-public sacreBLEU on the 0–100 scale with the canonical §3.2 donor registry (exact-normalized-text exclusion applied). All gaps are negative; these seeds enter only the within-implementation seed dispersion interval.

Seed	dev BLEU-4 [†]	REC	PURE	Gap
707	6.69	9.86	8.97	−0.89
808	7.30	9.87	8.38	−1.48
909	8.96	9.20	8.31	−0.89
1001	9.07	8.81	7.95	−0.87
1102	7.61	9.75	8.63	−1.12
1203	8.14	9.06	8.11	−0.95
1304	8.18	9.68	8.50	−1.18
1405	8.12	8.05	6.86	−1.19

[†]dev BLEU-4 is the training-time validation-selected value recorded by the original training protocol (legacy protocol; not comparable to the uniform full-pool free-decode dev BLEU-4 reported in main §4.2). REC, PURE, and Gap are canonical §3.2 beam-3 values.

3 Analysis Provenance and Multiplicity

Table S3 classifies every analysis in this manuscript by its role in the paper’s argument, its evidentiary status, and its multiplicity handling. The PURE–REC decoded-sacreBLEU gap under the released evaluator is the *selected probe gap* (post-selection descriptive; see main §4.1); every other analysis is secondary and is grouped by the role it plays—confirmatory control, descriptive triangulation, robustness check, or administrative record. C = confirmatory (frozen protocol with error control); D = descriptive (no formal error control).

Discovery timeline.

The +10.24 canonical reversal was first observed on 2026-07-24 in the materialization with exact-normalized-text exclusion (SHA-256 9170a530); an earlier pre-exclusion materialization (gap +11.01) had been noted on 2026-07-22 and survives only as a sensitivity analysis (Sup. N). The Jaccard-based source-conditioned retriever, PURE whole-donor-copy protocol, exclusion rule, and 641-query set were locked before the fourteen reconstruction training runs began; approximately four retrieval strategies and three construction variants had been explored on the released evaluator alone before the probe was frozen (multiplicity table in Sup. S). No external pre-registration exists for the specific +10.24 probe as a confirmatory hypothesis, so its CI is conditional on the post-hoc max-gap selection and is descriptive. The six primary reconstruction seeds (101–606)—specified before their results were inspected—constitute the only pre-specified evidence; the extension seeds, heterogeneous-family checkpoints, cross-evaluator re-decoding studies, distillation ladder, and extrapolation analyses

were added post-hoc. Analyses using the six-primary-seed set alone are labelled confirmatory (C); pooled or heterogeneous-checkpoint analyses are descriptive (D), as reflected in Table S3.

4 Checkpoint Registry

Table S4 is the definitive registry of every BT evaluator checkpoint used in this manuscript, with the analyses each family enters; all checkpoint counts in the text refer to these families.

Readout-competence association.

Thirty-eight constructible checkpoints plus the released evaluator have a uniform full-pool beam-3 free-decode readout (OR1 Sup. E protocol; the earlier 31-checkpoint set plus the eight seven-correction readouts). Table S5 reports the Spearman correlations over this set (`results/readout_gap_correlation.json`; `rescue_wd0` has a uniform readout but no canonical gap). Within the constructible family the readout is associated with decoded dev BLEU-4 ($\rho=0.86\text{--}0.88$, $n=17\text{--}18$), but the readout-gap association is *negative* ($\rho=-0.56$ excluding the released evaluator; -0.44 including it): better-reading-out constructible checkpoints have *more* negative gaps. The released evaluator is the single positive-gap checkpoint of the canonical (clip=1.0) family and lies outside the family trend on both axes (readout 0–12.08 vs. 78.8; train-minus-dev -0.7 to $+0.9$ vs. 65.4), so no within-family association supports either a tracking or an overfitting interpretation of its signature. These within-regime facts must not be extrapolated across clipping regimes: the grid of Sup. U shows the readout-gap relation turning positive once unclipped training enters the memorisation regime (readout 99.98, gap +11.19). Teacher-forced correlations over the 15 checkpoints with checkpoint-level stats: readout correlates negatively with train NLL ($\rho=-0.561$, $p=0.030$) and positively with train token accuracy ($\rho=0.509$, $p=0.052$).

Claims register.

Table S6 registers each headline claim with its estimand, unit of analysis, selection status, uncertainty quantification, and evidence tier, so that descriptive, confirmatory, item-level, and seed-level quantities are not read interchangeably (relocated from the main text; tiers as defined in main §3.7).

5 Donor-Pool Substitution Decomposition Details

This appendix gives the full path-sensitivity decomposition, factorial design, and balance checks referenced in §4.4.

Path sensitivity.

With SEEN-RAND640-MATCHED-v1 added, the systems form a 2×2 within-train design (pool size {7060, 640} $\times$ selection {max-Jaccard, Jaccard-matched}) plus the test-pool cell. Because corpus BLEU is non-linear in the support, all contrasts use the common 622-query support. Every ordering telescopes to the same total +14.7

[+13.3, +16.2], but the attribution varies by up to 7 BLEU points across orderings (Table S7). We report these as path-dependent descriptive contrasts, not mechanism decompositions.

Bootstrap family.

All decomposition CIs use 10,000 resamples. Four resampling designs (query-index paired bootstrap; three one-way cluster bootstraps for the origin contrast on template-family, SEEN-donor, and UNSEEN-donor clusters; and a joint multiway replicate) give origin-contrast CIs of [+7.2, +9.5] / [+7.1, +9.7] / [+7.1, +9.7] / [+7.7, +9.1], all matching the query-index CI [+7.3, +9.4] to within ± 0.3 , so dependence through shared donors is negligible.

Factorial design.

The 605-query factorial retains 605 of 641 targets (36 excluded for no joint match within tolerances; selection frozen before scoring). Table S8 reports the balance. Table S9 reports the full 2×2 cells, main effects, and interaction.

Multivariate balance and robustness.

On the 622-query common support, all six donor covariate SMDs between SEEN-PURE-MATCHED and UNSEEN-PURE are $|SMD| < 0.12$ (Table S10). Caliper sensitivity is small (tolerance 0.05/0.10/0.20 gives +8.2/+8.4/+8.6). Signer-strict matching (tolerance 0.10, $n=580$) attenuates the residual to +7.0 [+5.9, +8.1] — sign and significance unchanged. Inverse propensity weighting drives all four covariate SMDs to 0.00 and leaves the residual at +8.5 [+7.1, +10.1]. The path-based estimate is robust to observed-covariate balancing but remains an association, not an identified exposure effect.

6 Config-Faithful and Perturbation Details

Released config.yaml to training-code mapping (four-column, defect-corrected).

Table S11 maps each load-bearing field of the released `config.yaml` (verbatim copy shipped in `configs/released.yaml`, SHA-256 0321104a...) to three implementations of the semantics: *original framework* (the behaviour of the original training framework, established empirically from the released bundle itself—the logged `validations.txt` values, learning-rate decay pattern, and stopping point—or from the JoeyNMT 1.2 schema the config inherits), *our v1-v3 code* (the disclosed reconstruction attempts, including their defects), and *our faithful v4 code* (`train_faithful.py`, which corrects every identified defect). The forensic anchors for the original-framework column: (i) the released log’s step-14 **Translation Loss** 43,440.80078 equals 7,813 dev target tokens $\times \ln(\text{PPL } 259.84015)$, pinning `translation_normalization: batch` as token-sum CE divided by the number of sentences; (ii) all 20 logged learning-rate decays fire on the 6th consecutive non-improving validation, exactly `ReduceLROnPlateau(mode=max, factor=0.8, patience=5)` stepped once per validation on dev BLEU; (iii) the released run improved last at validation #130 (step

1820, BLEU 13.38) and then ran 72 further non-improving validations before ending at step 2828, so patience-5 *early stopping* is provably not the released stopping rule—the only stopping condition reproducible from the config is the learning-rate floor.

Epoch-validation approximation.

The released `config.yaml` specifies `validation_freq=14`, which JoeyNMT 1.2 interprets as 14 optimizer *steps* (≈ 28 steps/epoch means validating twice per epoch). Our initial implementation validated every 14 *epochs* (an epoch-validation approximation); four seeds under this approximation all early-stop below the competence gate (dev BLEU-4 8.6–9.5), show negative PURE–REC gaps (−1.1 to −0.2), and family-range pass-through (1.21–1.24) and holdout (7.4–7.6) — none reproduces the released signature (Table S12). Scaling capacity up (four larger variants: 4–6 layers, 384–512 hidden) overfits faster and reaches only dev BLEU 6.8–8.0. A long-schedule continuation (800 epochs, two seeds) pushes competence to a family maximum of 10.02 yet still shows no reversal (gap −0.45). The step-corrected re-run (Table S12) confirms the reconstruction plateaus below the released trajectory at matched step counts.

Perturbation battery.

Table S13 reports the full pose-perturbation battery. Reversing all frames collapses PURE from 23.79 to 7.72 (differential 16.1 [14.7, 17.5]; REC 5.5 [4.6, 6.5]); window-shuffling within one-second windows costs PURE 5.4 [4.4, 6.6]; 5% spatial jitter is free (−0.1); masking hand or face joints collapses both systems. The best rescue checkpoint (wd0, seed 202) loses only −4.3 under reversal, confirming it is far less sensitive to donor temporal order.

Released-checkpoint weight-noise perturbation (experiment A1).

The competence confound (released dev BLEU-4 13.38 vs. reconstructions ≤ 9.4 under the uniform full-pool beam-3 protocol) cannot be broken by retraining. Starting from the released weights, we add i.i.d. Gaussian noise to every parameter (excluding Batch-Norm running statistics) at scale $\sigma \in \{0, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ with two seeds each, and re-decode PHX-public REC/PURE and the dev set under the canonical donor registry (Fig. S1); small σ perturbs checkpoint identity while preserving competence.

The reversal is preserved under competence-preserving perturbation and collapses in lockstep with competence: at $\sigma \leq 3 \times 10^{-3}$ dev BLEU-4 stays at 12.50–13.38 and the gap at +9.50 to +10.35; the collapse happens only at $\sigma \geq 10^{-2}$ (dev 7.3–10.2, gap +3.2 to +4.2), and at $\sigma = 3 \times 10^{-2}$ both are extinguished together. The reversal is thus locally robust at competence parity and tracks competence rather than checkpoint identity. However, Gaussian noise probes only the isotropic weight-norm ball around the released checkpoint: it does not show that an independently trained, competence-matched model would exhibit the reversal.

Released-checkpoint fine-tuning (experiment A2).

Gaussian weight noise probes only the isotropic neighbourhood; to test local robustness along genuine gradient directions, we fine-tune the released checkpoint with small

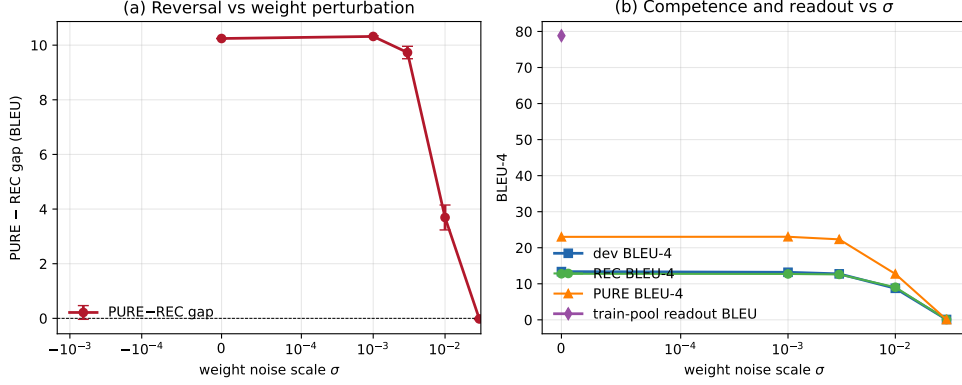


Fig. S1: Released-checkpoint weight perturbation (experiment A1). (a) PURE-REC gap vs. Gaussian weight-noise scale σ (two seeds per $\sigma > 0$; baseline at $\sigma = 0$). The reversal is preserved within ± 0.75 of $+10.24$ for $\sigma \leq 3 \times 10^{-3}$, where dev BLEU-4 stays within 12.50–13.38. (b) Dev BLEU-4, REC BLEU-4, PURE BLEU-4, and (at baseline) train-pool readout BLEU-4. The reversal and competence collapse together at $\sigma \geq 10^{-2}$; $\sigma = 10^{-1}$ produces an all-empty-decoding model and is omitted.

fixed learning rates (5e-6, 1e-5, 3e-5) for 60 epochs, 3 seeds each (9 runs total), using the identical training data and architecture as the released evaluator. Models are selected by best dev NLL and decoded under the canonical donor registry (beam-3). All fine-tuned variants preserve the reversal (gaps $+10.37$ to $+10.77$, all within ± 0.5 of baseline $+10.24$, dev BLEU-4 12.92–13.18, all above the reconstruction ceiling of 10.5 under the uniform protocol) and retain the training-pool readout signature (train BLEU 80.5–81.8, EM 72.6–74.0%, vs. baseline 78.8/70.7%). Results are in `results/released_finetune.json`. In contrast to isotropic noise, fine-tuning moves weights along data-driven gradient directions near the released optimum; the persistence of both the reversal and the readout signature under fine-tuning confirms they are not fragile singular-point properties. The same scope limit applies: like the Gaussian noise experiment, fine-tuning probes only the vicinity of the released weights and does not narrow the unobserved interval above the family ceiling (10.5).

Epoch-checkpoint decoupling search (experiment D).

We searched the four available intermediate epoch checkpoints (`rescue/seed_202_wd0` and `long_schedule/seed_202` at epochs 25 and 50, plus their best checkpoints) for a decoupling point that would break the within-family readout-gap coupling — e.g., elevated train-pool readout but PURE-REC gap still ≤ 0 . No such point exists: all six checkpoints stay in the (low readout, negative gap) quadrant, with train-pool BLEU 5.5–8.9 (1.6–2.0% EM) and PURE-REC gap -0.94 to -1.70 (per-checkpoint values in `results/epoch_decouple.json`). The within-family relationship between low readout and negative gap is therefore monotone across the entire observable trajectory.

7 Cluster-Aware Bootstrap for Headline and Reference Analyses

The headline gap (+10.24, canonical §3.2 registry) and the reference-attenuation result are reported with query-level paired bootstrap CIs. Because TN-PURE reuses donors and PHOENIX-2014T has template families and multiple signers, query-level resampling may underestimate uncertainty if dependence through shared donors, signers, or broadcast structure is non-negligible. Table S14 reports interpretable cluster-aware resampling designs alongside the query-level baseline, using precomputed per-item n-gram sufficient statistics for 10,000 resamples each.

The donor-cluster CI ([+8.87, +11.64]) matches the query-level CI ([+8.88, +11.62]) to within 0.03 BLEU and the donor two-way CI ([+8.79, +11.74]) to within 0.12, so donor reuse (61/565 donors reused, max 6×) does not materially inflate the item-level precision. The signer-cluster CI ([+9.29, +11.39], 9 clusters) is slightly tighter than the query-level CI, consistent with within-signer positive correlation; with only 9 clusters the percentile coverage of this cluster bootstrap is unreliable, so it serves as a sensitivity reference rather than a confirmatory interval (as flagged in the main-text Discussion). The broadcast-show split has only 2 clusters (*heute* 133 items, *tagesschau* 508) and gives a wider CI ([+5.16, +11.44]); we report it as an illustrative sensitivity check without confirmatory weight. The broadcast-date (297 clusters) and signer×show (17 clusters) CIs both match the query-level CI closely. All cluster designs are computed on the canonical materialization (the pre-exclusion version is reported as a sensitivity analysis in Sup. N). The donor-pool decomposition analysis of §4.4 additionally reports separate one-way cluster bootstraps and a joint multiway replicate for the origin contrast, all matching to within ± 0.3 (Appendix 5).

8 Per-Item Reference-Frame Sensitivity (matched 461-item subset)

The main-text reference-frame analysis (§4.3) and SI Fig. S5 report corpus-level attenuation under the matched 461-item confidence=1 subset. The per-item paired view of the same subset is reproduced here (Fig. S2): sentence-level sacreBLEU (0–100, same tokenizer/smoothing as corpus BLEU), released-evaluator canonical hypotheses. Panel (a) plots per-item scores under original vs. human references for REC (recorded test poses) and PURE (replay probe); PURE points sit high under the original reference and collapse toward the diagonal under the human reference, whereas REC points move less. Panel (b) plots the per-item paired gap (PURE–REC) under original (x) vs. human (y) references; the star marks the mean (sentence-level mean gap 8.11 $\rightarrow$ 0.09, consistent in direction with the corpus-level attenuation 10.03 $\rightarrow$ 0.95). Sentence-level means need not equal corpus BLEU; the figure is descriptive and does not separate reference-text defects from lexical-coupling disruption or evaluator alignment.

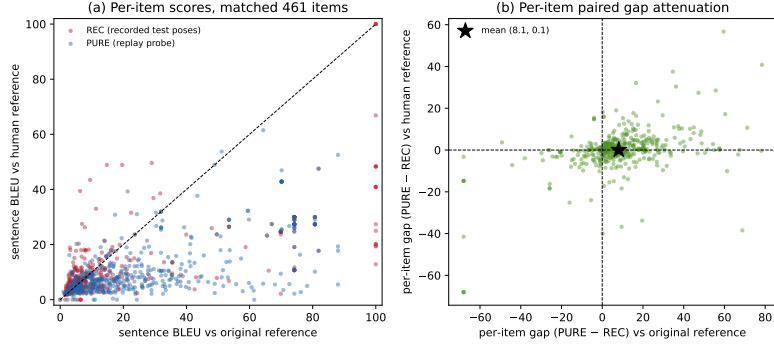


Fig. S2: Per-item reference-frame sensitivity on the matched 461-item confidence=1 subset (released evaluator, canonical hypotheses). See §8 for details.

9 Supplementary Oracle Diagnostics

Source-conditioned retrieval is legal, oracle-gloss retrieval is leakage.

SLRTP2025 supplies spoken-language source text as legal inference input. Under pure whole-donor copy, source-text Jaccard scores 23.02 sacreBLEU (0–100 scale, as throughout this appendix), LaBSE 19.2, and multi-SBERT 16.5; the PT-composed source-text condition scores 18.86 (see main text, Table S47). Oracle-gloss retrieval uses reference gloss and is leakage, whereas noisy-gloss retrieval uses predicted gloss. Thus the strongest legal source-conditioned result is the 23.02 whole-donor stress test, which uses no generator or scaffold.

Disclosure of selection/evaluation reuse.

The white-box oracle analyses in this appendix reuse the same target reference and evaluator for both donor selection and decoded evaluation. This is an intentional adversarial design: diagonal entries quantify susceptibility to a selector that knows its evaluator, not an independent validation of retrieval quality.

Cross-checkpoint transfer matrix.

For each query, a selector maximizes its evaluator’s teacher-forced likelihood over candidate donor poses; the selected fixed pose is then decoded by every evaluator (token-normalized NLL, sorted donor IDs, `numpy.argmax` first tie-break). All seven diagonal margins are positive (mean +0.0150, 95% paired-bootstrap CI [+0.0105, +0.0198]). Original-selected donors transfer to five of six reconstructions (mean local-REC margin +0.0074, [+0.0016, +0.0135]); the reverse direction fails in all six cells (reconstruction-selected donors evaluated by Original, mean margin −0.0455 [−0.0554, −0.0358]). The full 7×7 matrix, unrounded cells, margins, and CIs are in the artifact.

Candidate-pool size.

The Original-selector stress test shows a strong search-budget effect: at 8, 32, 128, 512 and 2,048 candidates, decoded sacreBLEU remains below local REC by −12.1,

−9.2, −5.7, −3.6 and −1.3 BLEU points respectively; only the full 7,060-donor pool is detectably positive (+1.7, [+0.4, +3.0]).

Reference-overlap exclusion.

Masking 1,036 exact-text query–donor pairs before selection: Original-on-Original decoded BLEU is 13.86 versus local REC 12.78 (absolute difference +1.08 BLEU points), but the paired relative-margin 95% CI crosses zero; the frozen joint likelihood-and-interval success rule is met in 0/7 checkpoints. Masking all 19,694 pairs with source-token Jaccard ≥ 0.5 : the score falls to 8.30 (−4.47), again 0/7 meet the rule. The full-pool diagonal advantage does not survive as a statistically robust decoded-BLEU result after exact-text exclusion and disappears after high-overlap exclusion.

Noisy-gloss retrieval is split-dependent.

A 6-layer text→gloss Transformer (dev BLEU-4 13.17) yields noisy-gloss retrieval sacreBLEU 12.7 on PHX-public (strong-NMT), versus 1.6 for a weak NMT (BLEU-4 1.15) and 6.1 for XLM-R embedding retrieval; donor exclusion reduces the strong-NMT value to 10.9 at $\tau=0.5$, and the same control reaches 0.4 on the template-disjoint holdout (Table S18).

PHX-public pure-donor donor-exclusion τ -sweep.

To test whether the headline reversal survives donor exclusion on the same split, we applied a source-text Jaccard τ -sweep and exact-text-exclusion to the canonical pure text-nearest donor on PHX-public under the original SLRTP2025 BT (Table S15). The reversal survives exact-text exclusion (23.0 vs REC 12.8, 32/641 donors replaced) and source-Jaccard exclusion up to $\tau=0.5$ (17.8 vs REC 12.8); only at $\tau=0.3$ and $\tau=0.1$ does the retrieval-vs-REC gap turn negative.

As a same-paradigm proxy for the SLRTP2025 winner (whose checkpoints are not public), the noisy-gloss retrieval+LM system shows the same pattern: it survives exclusion up to $\tau=0.5$ (10.9, still $6.8\times$ the PT-only baseline and $11.7\times$ the random-donor mean; mean selected-donor Jaccard 0.46) and collapses only at $\tau=0.1$.

10 Template-Density, Holdout Boundary, and Donor-Exclusion Analyses

Template density does not gate the reversal.

Stratifying the 641 test sequences by template density (gloss bigram-Jaccard with nearest train donor), retrieval exceeds REC at all levels: low +10.7, mid +10.1, high +12.4 BLEU points (Table S16). Density moderates magnitude but does not gate presence.

Holdout boundary controls (historical analyses).

The controls in this paragraph predate the host-storage incident of 2026-07-14; their per-item decode artifacts are retained in the artifact repository, but the training runs themselves could not be recovered from disk, so they cannot be re-executed from the

public bundle and are labelled historical. Truncating to 64/32 frames shrinks the gap from +10.24 (canonical; +11.0 in the pre-exclusion materialization) to +10.5/+5.6 but the reversal persists. Four holdout controls (Table S17) all fail to reproduce the reversal: the same-evaluator seen-holdout gives REC 77.0 (readout), BT-retrained on unseen holdout gives gap -0.9 , retrained BT on PHX-public gives gap -0.4 , and cross-fit A/B gives gap -0.1 .

Frozen template-holdout with retrained components.

A deterministic template-holdout (14,988 train-core windows, 1,538 holdout windows, disjoint by window, source sequence, and exact gloss template) with all components retrained on train-core shows text-nearest retrieval above random (+2.3) but far below REC (-10.8): the shortcut is real over random but does not exceed REC on the clean split. Donor-exclusion sweeps show the advantage persisting at $\tau=0.5$ and collapsing only at $\tau=0.1$ (Table S18).

The SLRTP2025 test vocabulary is nearly closed (95.4% gloss types in train); nearest donors share mean gloss unigram containment 0.557 and bigram-Jaccard 0.302, but Progressive Transformer outputs show no donor-proximity effect ($\rho=-0.06$, $p=0.12$), versus +0.29 for retrieval.

11 Transcript-Slot Diagnostic Details

This appendix documents the rule-based German weather slot extractor referenced in the main text. The extractor identifies four slot families (number/temperature, time, place, weather-event) using a frozen lexicon and normalization rules. On a 50-reference audit (seed 20260729), per-family precision was 0.91–1.00 and recall 0.75–1.00 (overall precision 0.97, recall 0.86); errors are dominated by recall misses from inflections, compounds, and lexicon gaps. A stratified audit of 50 decoded hypotheses (25 REC-decode, 25 PURE-decode, seed 20260730) showed precision nearly equal (0.96 vs 0.97) but recall lower on PURE-decode (0.83 vs 0.91), because donor-transcript-style outputs use more inflected verbs and compounds (*regnet*, *unwetterwarnungen*) that the lexicon misses. The measured PURE slot-F1 is thus attenuated more than REC’s, so the PURE>REC contrast is, if anything, underestimated. The audit used a single auditor; inter-annotator agreement is unavailable. The full lexicon, normalization rules, and per-family precision/recall are in the artifact.

12 Competence Extrapolation Details (Exploratory)

This appendix gives the full OLS/GP extrapolation analysis referenced in the main text (RQ1). We emphasize that this is exploratory and the prediction intervals are descriptive visual guides, not calibrated inferential tests, because the 65 unique non-released checkpoints share data, code, and hyperparameters (non-IID) and the target point (dev BLEU-4 = 13.38) lies outside the observed range.

Method.

We fit the gap–competence dose–response over the 65 unique non-released weight binaries (the 66 decoded runs reduce to 65 after SHA-256 deduplication of the validation-freq-misread/long-schedule seed-202 collision), spanning dev BLEU-4 0–10.5 (61 non-degenerate checkpoints; five degenerate checkpoints sit at dev ≈ 0), using three models: OLS linear, OLS quadratic, and a Gaussian Process with RBF + white kernel. Each model predicts the PURE–REC gap at dev BLEU-4 = 13.38.

Results.

Table S19 reports the predictions and 95% PIs at dev BLEU-4 = 13.38. All predictions are far below the observed canonical +10.24. Table S20 reports a leave-one-family-out cross-validation; the OLS quadratic prediction is the most sensitive (shifts from -1.09 on all 66 checkpoints to +2.88 when restricted to the 61 non-degenerate checkpoints with dev ≥ 4.5 , and to +0.23 when the distillation-family checkpoints are removed), while GP predictions stay negative with leave-one-family-out upper bounds below +0.3.

Caveats.

First, the GP upper bound (+0.3) partly reflects RBF kernel prior shrinkage toward the mean in the extrapolation region (sparse data beyond dev BLEU-4 10.5), not purely data evidence. Second, the OLS quadratic fit is sensitive to the distillation family and to the inclusion of the degenerate checkpoints. Third, the 61 non-degenerate checkpoints are not IID. We therefore report these fits only as descriptive visualizations and explicitly avoid claiming that “competence alone is unlikely to explain the reversal” on the basis of these fits. The unobserved (10.5, 13.38) interval could in principle harbor non-linear behavior that the present experiments cannot rule out.

Per-item gap decomposition (experiment C).

We regress the per-item sentence-level PURE–REC gap (sacreBLEU sentence BLEU-4 under the released evaluator, canonical donor registry; mean 8.4, SD 18.7, $n = 641$) on interpretable item-level features: donor–reference token-set Jaccard, normalized character Levenshtein, an exact-match-train-text flag, template density (training texts sharing the reference’s 60-character prefix), high-overlap-train (training texts with Jaccard ≥ 0.5), reference length, donor reuse, a confidence=1 flag, and a 9-level signer block (Fig. S3). The full model explains $R^2 = 0.33$; the strongest leave-one-out contributors are high-overlap-train (0.093), template density (0.051), and donor–reference Levenshtein (0.039); the signer block contributes 0.008—signer identity is essentially irrelevant.

The standardized-coefficient signs are informative: items where the donor is lexically close to the reference (Levenshtein similarity, $\beta^* = +6.1$; Jaccard, $\beta^* = +2.9$) show *larger* gaps, whereas items with many high-overlap training neighbors ($\beta^* = -7.6$) or high template density ($\beta^* = -7.3$) show *smaller* gaps—the reversal is largest where retrieval delivers a donor matching the reference register closely but the recorded poses do not independently recover that text, precisely the lexical-coupling mechanism predicted by the construction. The unexplained 67% of variance is consistent with

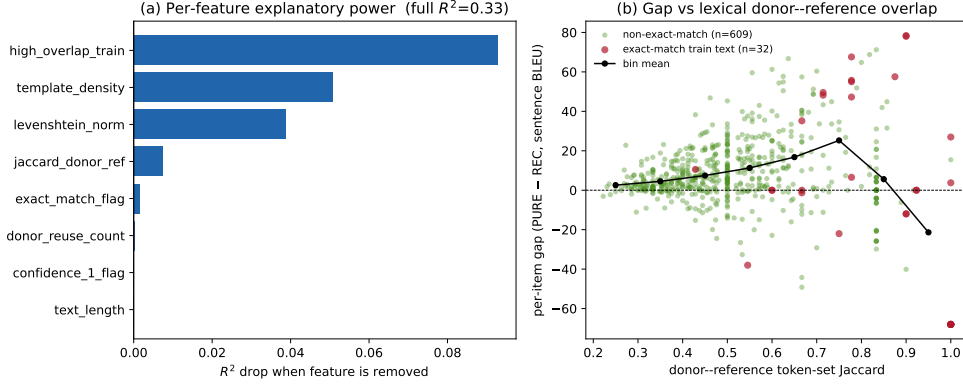


Fig. S3: Per-item gap decomposition (experiment C). (a) Per-feature R^2 drop when the feature is removed from the full OLS model ($R^2 = 0.33$, $n = 641$). High-overlap-train count and template density are the strongest single contributors; the full signer block contributes $\leq 1\%$. (b) Per-item gap vs donor-reference token-set Jaccard; red marks the 32 items whose reference text exactly matches a training text, green the remaining 609; the black line is the bin mean. The gap grows with donor-reference lexical overlap, consistent with the lexical-coupling mechanism. Outcome is sentence-level BLEU-4 (sacreBLEU 2.5.1, 13a, exp smooth); sentence-level mean need not equal corpus BLEU.

per-item pose variability and decoder stochasticity; the decomposition localizes the reversal to a constructible lexical-coupling property of PHX-public’s template-dense register, consistent with Alkain et al. (2026).

13 Supplementary CSL-Daily Boundary Control

CSL-Daily (Chinese Sign Language) serves as a non-template-domain control, but its recognizer is too weak for robust claims. Under Mode B (1,176 sequences), text-nearest donor copy reaches only 1.6 sacreBLEU vs REC 26.7 (Table S21). A readout-ratio check (64 train + 64 test windows) gives $\text{train} \approx \text{test}$ BLEU-1 (0.0019 vs 0.0021, ratio ~ 0.9), far below PHX’s 6.1: the readout signature does not generalize beyond the released PHX evaluator.

14 Leakage Sanity Checks for Train-Pool Readout

This appendix documents the five sanity checks referenced in §4.2 that rule out implementation-level artifacts as explanations for the released evaluator’s 78.8 BLEU / 70.7% EM train-pool readout.

Experiment A: Code-path audit.

The `back.translate()` function in `src/models/back_translate.py` (lines 362–426) calls `model.run_batch(translation_beam_size=3, translation_beam_alpha=-1)`,

which invokes `bt_search.beam_search()`. The decoder input is seeded with `<bos>` only and extended autoregressively with the model’s own previously generated tokens. The reference text is used exclusively in the post-hoc BLEU/EM scoring step; it never enters the model forward pass. We verified this by code inspection and by confirming that the same model loaded in a fresh Python process produces identical outputs.

Experiment B: Pose-text permutation.

We decoded the first 1,000 training items with beam=3 (BLEU 78.61, EM 70.1%), then permuted the reference labels across items to break pose-text correspondence. Three random permutations yield:

Experiment B': Input-side pose permutation with output-equivariance.

The label permutation above shuffles reference labels *after* decoding, so it cannot exclude ID-based or cache-based retrieval (a cache lookup by ID would also produce low BLEU when labels are shuffled). To properly test pose conditioning, we held item IDs and text metadata fixed, shuffled the *pose tensors* across 100 training items *before decoding*, and performed output-equivariance verification. For all 100 items, the hypothesis produced when pose_j is decoded in slot i ’s position exactly matches the hypothesis originally produced for pose_j in its own slot j (100% exact string match). Output length follows the pose tensor, not the slot ID (100% donor-length match vs. 5% slot-length match). This rules out a pose-ignorant pure ID/cache lookup under the tested decoding path and confirms sensitivity to the substituted pose tensor. Additional degenerate-pose controls (zero, Gaussian noise, single repeated pose for 200 items) all collapse BLEU to <1 , and reversed batch order reproduces the original BLEU exactly, confirming per-item determinism.

Experiment C: Determinism.

Beam search with no stochastic sampling produces deterministic outputs. We record SHA-256 hashes of all per-item decoded hypotheses in the artifact for cross-process verification.

Experiment D: Stratified EM.

Exact-match rate is high and broadly distributed, not concentrated in a single subgroup: Even unique training texts (appearing exactly once) achieve 70.2% EM, confirming the readout is not driven by template repetition alone. Of the 7,060 hypotheses, 74.8% match some training text exactly (70% match exactly one—their correct paired reference).

Experiment E: Same-signer held-out control.

We decoded 500 test-split poses (unseen during training) from the same 9 signers present in the training pool. This is a same-signer held-out control, not a fully matched membership control: the test items are not matched on length, text-similarity, broadcast date, or template density, so the train-test gap reflects a combination of training-pool exposure and distributional shift. The same signers appear in both splits. The 67-point BLEU gap between seen and unseen items is consistent with

training-pool exposure contributing to the readout, but without full covariate matching (length, text-similarity, template density), we cannot attribute the gap solely to exposure vs. distributional shift.

15 Pre-Exclusion Materialization Sensitivity

The canonical §3.2 donor registry applies exact-normalized-text exclusion. An earlier frozen materialization predating this exclusion gives PURE 23.79 (gap +11.01) instead of the canonical 23.02 (gap +10.24). We retain it only as a sensitivity analysis.

Per-item provenance.

On 610/641 items (95.2%) both materializations produce identical donor choices. On the 31 differing items, a post-hoc candidate search found that 30 had at least one excluded Jaccard=1.0 candidate whose released-evaluator decode matched the earlier hypothesis, consistent with the exclusion rule explaining the difference on those items; one item remained unresolved (`results/donor_identification_31.json`).

14-reconstruction sensitivity.

Re-decoding all 14 reconstruction checkpoints under both materializations gives mean $|\Delta_{\text{gap}}| = 0.65$ BLEU (max 1.45). The sign of the gap is unchanged on every checkpoint, and the 14-seed within-implementation dispersion interval remains strictly negative under both materializations, so the non-replication conclusion is unaffected. All headline numbers in the main text and in Sup. G use the canonical materialization (SHA-256 9170a53026ab3263b451a3632a9e02318745acabf12f0b679921477afbafa301); this appendix is the only place the pre-exclusion values appear.

16 BLEURT-Substitute Validation

Canonical BLEURT-20 ships a TensorFlow checkpoint that does not load cleanly in our PyTorch-only environment. As a non-standard substitute we use `Elron/bleurt-large-512` for one secondary reference-sensitivity check only (the BLEURT row of the reference-frame analysis). It is not used for any primary claim, table, or figure, and is not described as equivalent to BLEURT-20.

Validation.

On 100 German sentence pairs we scored both canonical BLEURT-20 (via the reference TensorFlow runtime in an isolated CPU container) and the substitute. The substitute has Spearman rank correlation $r = 0.87$ against canonical BLEURT-20 and a mean signed bias of -0.47 (substitute scores lower on average). The bias is roughly uniform across score levels and does not change the qualitative reference-sensitivity conclusion (BLEURT is less sensitive to reference replacement than BLEU, consistent with Czehmann et al. Czehmann et al. (2026)). Full per-pair scores, the container hash, and the substitute model revision are in the artifact.

Scope limit.

Because the substitute is not BLEURT-20, the BLEURT row is demoted from the main metric panel to this appendix. Readers should treat the absolute BLEURT-substitute values as approximate; only the directional reference-sensitivity finding is reported.

17 Additional Protocol Details Moved from the Main Text (Sup. Q)

Text-nearest tie-breaking.

Jaccard ties are resolved by minimum character-level Levenshtein distance and then by minimum SHA-256 hash of the training key. The retrieval algorithm also excludes donors whose NFKC-normalized text equals the query’s (exact-normalized-text exclusion).

Scorer equivalence.

The vendored official scorer (SacreBLEU 1.4.2) and the paper scorer (SacreBLEU 2.5.1, `tok:13a`, `smooth:exp`) give identical corpus BLEU on all 28 evaluation cells (maximum $|\Delta| = 0.0000$; byte-identical hypotheses, all n-gram orders non-zero). The main text reports the 2.5.1/13a signature as primary.

Metric implementations.

All secondary metrics use the official SLRTP2025 implementations: they reproduce the official repository values (BLEU-4 12.777, chrF 34.585, WER 85.775, ROUGE 35.196 Walsh et al. (2025)) exactly. Implementation choices matter for chrF (corpus-level sacrebleu `CHRF` defaults; a sentence-level variant yields 48.05 instead of 34.59) and WER (jiwer 3.1.0 with official normalization; raw whitespace-split yields 79.26 instead of 85.78). BLEU-4 is invariant; the official implementations are fixed throughout.

Audit design overview (moved from the main text).

The main-text audit design figure (SI Fig. S4) fixes a set of pose outputs and scores them under two contrast axes: the evaluator checkpoint (released, fourteen recipe reconstructions, heterogeneous diagnostic checkpoints) and the reference frame (speech-transcribed vs. human sign-to-text).

18 Main-Text Tables (Moved from the Main Text; Sup. P)

The tables in this section were moved from the main text to satisfy the journal page limit; the main text references them as SI Table S22–S33.

19 Reconstruction Sanity Checks

This appendix documents the three sanity checks that rule out a systematic implementation defect shared across all 68 training runs. Scripts: `scripts/e_overfit_test.py`,



Fig. S4: Audit design: a fixed set of pose outputs is scored under two contrast axes. (a) Data and provenance: the official PHOENIX-2014T split (7,096/519/642) was materialized by the SLRTP2025 release as 7,060/515/641 poses, matching the released evaluator’s training manifest (36/4/1 official records absent). (b) Fixed output conditions for one German source sentence: recorded test poses (REC), Progressive-Transformer output (PT), whole-donor copy TN-PURE (source-text token-set Jacard, exact-normalized-text exclusion), and PT-scaffold composition TN-PTCOMP. (c) Two contrast axes: the evaluator (released, fourteen recipe reconstructions, heterogeneous diagnostic checkpoints) and the reference frame (speech-transcribed vs. human sign-to-text), with pose outputs and the donor registry fixed; the primary estimand is $\Delta_c = \text{BLEU}_c(\text{PURE}) - \text{BLEU}_c(\text{REC})$.

scripts/e_checkpoint_stats.py, scripts/e_inference_regression.py.
Results: results/overfit_test.json, results/checkpoint_stats.json,
results/inference_regression.json.

Small-sample overfit test.

A reconstruction model was trained on 50 (and separately 100) randomly sampled training sequences for ≈ 200 epochs using the identical pipeline as the paper-derived reconstruction runs. On the 50-sample set, teacher-forced NLL fell from 7.66 to

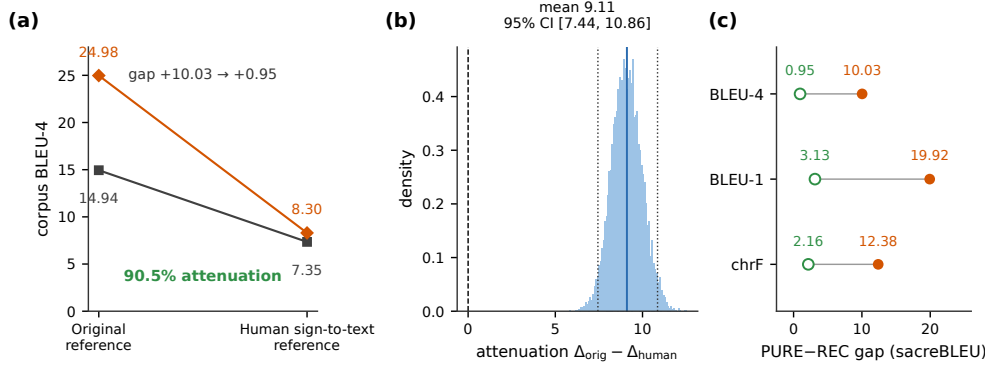


Fig. S5: Human sign-to-text references remove most of the replay advantage on the matched high-confidence subset ($n = 461$; released evaluator, canonical hypotheses). (a) Corpus BLEU-4 under original vs. human references: REC 14.94 $\rightarrow$ 7.35, PURE 24.98 $\rightarrow$ 8.30; the reference-frame contrast reduces the gap +10.03 $\rightarrow$ +0.95 (90.5%). (b) Bootstrap distribution of the contrast (10,000 paired resamples; mean 9.08, 95% CI [7.44, 10.86]). (c) PURE-REC gap under original vs. human references across BLEU-4, BLEU-1, and chrF. References are back-translations by a single deaf fluent L2 DGS signer Czehmann et al. (2026); replacement cannot distinguish reference-text defects, paraphrase shift, BLEU floor effects, or disruption of the lexical coupling.

0.009 and free-decode sacreBLEU-4 reached 100.00 under beam-3 decoding. On the 100-sample set, free-decode BLEU also reached 100.00 (300 epochs). The reconstruction code can therefore overfit small data to near-perfect free-decode BLEU, ruling out a fundamental defect in checkpoint loading, tokenizer, padding-mask, EOS, loss normalization, or the beam-search decoding path.

Unified inference regression.

Released and reconstruction checkpoints are loaded through the identical `make_back_translation_model()` function and decoded with the identical `back_translate()` pipeline. All 15 checkpoints (released + 14 reconstructions) share byte-identical `txt.vocab` and `gls.vocab` files (SHA-256 confirmed). The released checkpoint decoded through this path reproduces the canonical REC value exactly (12.78 on 641 PHX-public items). The tested reconstruction seeds (101, 202, 303) decode REC at 9.24, 8.22, 9.10 respectively, matching the main-text table. Inference code path is identical for all checkpoints.

Teacher-forced vs. free-decode diagnostics.

Table S46 reports per-checkpoint teacher-forced token accuracy, NLL per token, and free-decode BLEU. Teacher-forced values are on 500 randomly sampled training items; free-decode values are full-pool (7,060 train / 515 dev) under the uniform beam-3 protocol (the per-seed training-time free-decode values in

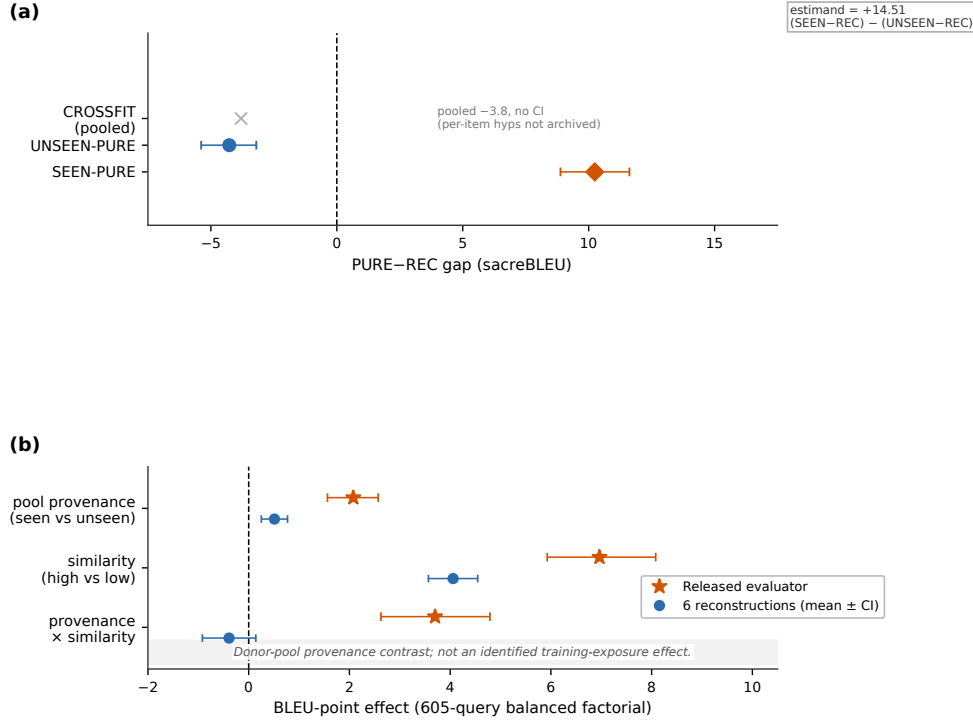


Fig. S6: The replay advantage co-occurs with training-pool donor provenance, but provenance and similarity remain confounded. (a) Donor-construction gaps relative to local REC: SEEN-PURE +10.24; UNSEEN-PURE -4.27 (paired-item CI [-5.39, -3.20]); the pool-substitution estimand $(\text{SEEN} - \text{REC}) - (\text{UNSEEN} - \text{REC}) = +14.51$; CROSSFIT pooled -3.8 (text only, no CI). (b) Balanced 605-query factorial: released evaluator (star) vs. six-reconstruction means (circles) for the pool-provenance and similarity main effects and their interaction. Donor-pool provenance contrast; not an identified training-exposure effect.

`results/checkpoint_stats.json` were computed on a 500-item subset with a length-50 cap and are not comparable).

Key observation: the released evaluator achieves 96.5% teacher-forced token accuracy on training data versus 55–60% for reconstructions. On the development set, reconstructions achieve slightly lower teacher-forced NLL (2.63–2.68 vs. released 3.24, consistent with less overfitting) but comparable token accuracy (45.8–47.4% vs. 48.6%). The gap in free-decode BLEU is visible already in teacher-forced training-set metrics, not solely a decoding phenomenon.

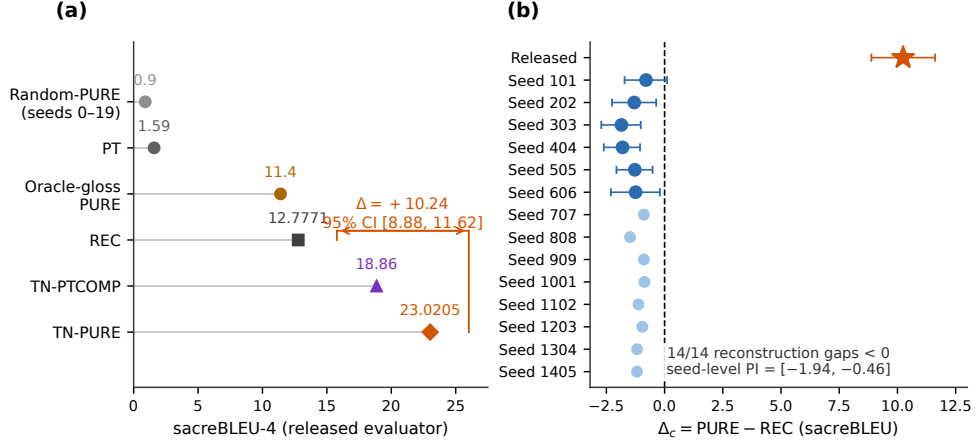


Fig. S7: Adversarial unit-test response under the released evaluator and its absence across the canonical (clip=1.0) panel (relocated from the main text). (a) System scores (sacreBLEU-4, 641 queries): TN-PURE 23.02 vs. REC 12.78 ($\Delta = +10.24$, post-selected), with TN-PTCOMP 18.86, PT 1.59, oracle-gloss 11.4, random-donor 0.90 (± 1 SD, 20 seeds). (b) Per-checkpoint gaps: released (+10.24) vs. fourteen legacy reconstructions, all negative (donor-cluster bootstrap CIs; dispersion $[-1.85, -0.80]$). The remaining families (full panel $[-2.80, -0.21]$) are in main Fig. 2; none is positive—an absence the main text (§??) traces to the shared clip threshold (unclipped: +11.19).

20 Dev-Split Consistency Check and Reference-Free Retrieval

Dev-split consistency check for the headline probe.

The canonical probe panel was constructed and explored on the 641-item PHX-public *test* split; the probe response was therefore re-checked on the development split (515 items), which was never used for probe construction or exploration. The identical TN-PURE-v1 construction (NFKC normalization, token-set Jaccard, character-Levenshtein then SHA-256 tie-breaks, exact-normalized-text exclusion, donors from the same 7,060-item training pool) was applied with dev items as queries, and both REC (recorded dev poses) and PURE (donor poses) were decoded under the released evaluator with the canonical beam-3 protocol. The released evaluator’s dev REC BLEU-4 is 13.379, matching the 13.38 logged in its released `validations.txt` to three digits and independently validating the decode pipeline. The dev-split PURE score is 21.47, a gap of +8.10 versus the test-split +10.24 (REC 13.38 / PURE 21.47 vs. 12.78 / 23.02). The reversal therefore reproduces, with consistent direction and comparable magnitude, on a split untouched by probe exploration. This is a *secondary consistency check*, not an independent held-out confirmation: the dev split is used for checkpoint selection and LR scheduling in every reconstructed run, and 13.38 is itself the released run’s selection metric, so the two splits are not exchangeable evidence. Per-item files and the dev donor registry (SHA-256

recorded in the result JSON) are in `results/dev_split_confirmation.json` and `results/dev_split_confirmation.items/`.

Pose-retrieval probe: donor selection without the scored reference.

The text-based retrievers (Jaccard, LaBSE, multi-SBERT) all select donors using the query’s text—which is identical to the BLEU scoring reference (target leakage by construction). To sever this channel while retaining a meaningful retrieval signal, donors were selected by *pose* similarity to the query’s own recorded pose sequence, an input-side signal available to a deployed system: each sequence (after the canonical 12.5 fps subsampling) is summarised by its time-averaged pose and mean absolute frame-to-frame velocity, each part z-normalized per dimension over the pooled queries and donors and L2-normalized, and the donor minimising the combined Euclidean distance is selected (ties by donor-ID hash, the same final tie-break convention as the canonical retriever; no text is used anywhere in the loop). Under the released evaluator on the same 641-item test split, POSE-PURE scores 0.45 (gap -12.33 vs. REC 12.78), indistinguishable from the random-donor baseline (0.90, gap -11.88) and far below text-retrieval TN-PURE (23.02, $+10.24$). The positive probe response is therefore *entirely dependent* on selecting donors with the scored reference text: any reference-free selection rule examined (random, pose-similarity) collapses to deeply negative gaps. Implementation and per-item files: `scripts/e_pose_retrieval_probe.py`, `results/pose_retrieval_probe.json`, `results/pose_retrieval_probe.items/`.

Table S47 extends the main-text multiplicity summary with the pose-signature row: reference-free retrieval joins the random baselines far below REC.

21 Retrieval-Query Decoupling Grid

Question answered: does the released evaluator’s positive probe response require selecting donors with the *scored reference text* specifically, or does any lexically competent text-based selection produce it?

Design. A 2×2 grid crosses the retrieval query text (original reference vs. Czehmann et al. human back-translation) with the scoring frame (original vs. human), on the matched 461-item confidence=1 subset. Retrieval is the same source-Jaccard argmax with the canonical exact-normalized-text exclusion (the excluded text is the query’s original reference in both arms); no re-decoding is needed because the released evaluator’s hypotheses for all 7,060 training-pool poses are cached (`results/full_readout/backTranslation_PhiX_model.json`). Outputs: `results/retrieval_query_decoupling.json` (`scripts/e_retrieval_query_decoupling.py`).

Three readings. (i) *Replacing the query alone removes the response* (-4.02) although the selected donors remain lexically close to the original references (Jaccard 0.30) and decode far above the random-donor regime, so the collapse is not explained by retrieval quality alone. (ii) *Re-coupling query and scoring frame on the human back-translations partially restores the response* ($+4.29$), at a magnitude consistent with the weaker lexical coupling of that diagonal (donor–human-text Jaccard 0.33 vs. 0.55

on the original diagonal). (iii) Uniform randomisation within the Jaccard top- k (query = original, full 641 queries) interpolates between the regimes: gap +7.06 at $k=5$ and +3.34 at $k=20$ (10 draws each), against -11.87 for uniform random donors. Together with the text-free selectors of Sup. S (pose-signature -12.33), the response tracks the lexical identity of the retrieval query with the scoring reference rather than text-based selection as such; all quantities remain descriptive associations under a single evaluator, and the human frame is a single-annotator resource.

22 Gradient-Clipping and CTC-Normalisation Sensitivity (Sup. U)

Question answered: are the training-pool readout signature and the positive probe response unrecoverable properties of the released evaluator, or artefacts of the two disclosed training-dynamics inferences (gradient-clip threshold; CTC-term normalisation)?

Design. Seed-42 sensitivity runs of the seven-correction protocol (`src/training/train_faithful.py`) varying one knob each: CTC denominator (sentences / gloss tokens / input frames / raw sum) at clip 1.0; gradient-clip `max_norm` (1.0 = the faithful run, 2.0, 3.0, 5.0, none = ∞); and a candidate-upstream run (no clipping + CTC raw sum, the literal `_train_batch` semantics of the underlying framework). All runs use the released seed-42 init, the released config otherwise, and the `--stop-frozen` termination (LR frozen at the plateau scheduler’s numerical floor with the best checkpoint unchanged); none belongs to the canonical 76-run panel (registry directories `checkpoints/ctc_clip_sens/`, `checkpoints/faithful_steps/`). Evaluation: canonical 641-query REC/PURE protocol, full 7,060-item training-pool free-decode readout with exact-match, via `scripts/eval_step_ckpts.py` into `results/faithful_steps_eval.json`.

Source evidence for no clipping. The released repository reuses a signjoey-derived pipeline: its `validations.txt` column layout matches signjoey’s `_add_report` line format field-for-field, and the data/model code paths correspond. The candidate upstream is the public repository `NaVi-start/Sign-IDD-SLT` (Apache-2.0), whose main HEAD is commit `249d3cd8fc249b1a06eba39f84cb2d289ed37bce` (2025-04-27, “Update README.md”, author HJY)—a date consistent with the SLRTP2025 timeline. It is an independent public repository, not a GitHub fork of `neccam/slt` (the original Sign Language Transformers repository, whose public history ends in 2020); the correspondence is at code level. The commit is verifiable against the repository HEAD at clone time, and a verbatim source archive (code, configs, licence; PHOENIX-derived annotation pickles excluded for licensing) with per-file SHA-256 hashes ships in the artifact at `third_party/signjoey-signidd-slt-249d3cd8fc2/` (`PROVENANCE.json`, `SHA256SUMS`). In that source’s `signjoey/builders.py`, `build_gradient_clipper` inspects the config fields `clip_grad_val` and `clip_grad_norm`: the former selects value clipping (`clip_grad_value_`), the latter norm clipping (`clip_grad_norm_`), the two are mutually exclusive, and when neither is present the builder returns `None`—no clipping at all. The released `config.yaml` contains neither field, so under the candidate upstream semantics the released configuration implies no clipping; this

reading is conditional on the framework version and unconfirmed by any released training transcript. The same source pins the CTC-term semantics: `_train_batch` divides the translation loss by `num_seqs` (the `batch` normalization) but the recognition loss only by `batch_multiplier` (`=1`), i.e. the raw CTC sum—the source carries an explicit developer comment (`# TODO (Cihan): Add Gloss Token normalization (?)`) acknowledging the missing normalisation. Our `max_norm=1.0` was inherited from the `v1-v3` scripts and disclosed as an author inference; this table overturns it. The inference chain is source-semantic (framework version assumption), not a released transcript; the behavioural confirmation is the “none” row reproducing both previously unreproduced properties. (The source’s stock stopping logic—stop when a plateau decay fires and no improvement occurred at the pre-decay LR—would have terminated the released run at roughly the second post-best decay, a dozen validations after its step-1,820 best; the released run instead logged 72 further validations, so the release’s own stopping rule remains externally terminated and the step-matched analysis of the main text applies.)

Unclipped trajectory. The unclipped seed-42 run improves dev BLEU-4 monotonically to 15.69 at the released run’s own best step (1,820), reaches its best 16.32 at step 2,352, and continues to the scheduler’s numerical floor (step 7,014, 501 validations, LR frozen at $4.4\text{e-}8$) without the best checkpoint changing; it was then terminated externally (the same manual rule as the published seven-correction runs). Pre-clip gradient norms over the first ten optimiser steps (377, 568, 145, 33, ...) exceed the 1.0 threshold by up to two orders of magnitude, so the clipped and unclipped regimes diverge from the first update: at clip 1.0 every step is direction-preserving truncation, and the memorisation regime is never entered.

Step-matched unclipped readout. A fresh unclipped run with checkpoints fixed at the released run’s own best (1,820) and final (2,828) steps shows the memorisation regime is entered *early*: at step 1,820 the readout is already 99.94 BLEU / 99.7% EM (dev 15.69, REC 14.64, PURE 26.32, gap +11.68); at step 2,828, 99.98 / 99.9% (dev 15.80, gap +11.49). The overshoot relative to the released evaluator’s step-1,820 readout of 78.8 is therefore not an artefact of training longer than the released run: under unclipped dynamics the training pool is essentially memorised before the released run’s best checkpoint.

Multi-seed confirmation. Across seeds 42–44 the unclipped protocol gives dev 15.74–16.32, REC 14.32–15.96, gaps +11.19/+11.31/+10.31, and readouts 99.98/95.25/99.98 BLEU (90.5–99.9% EM): the reproduction of both properties is stable across seeds, and every seed’s gap is within 1.1 BLEU points of the released +10.24 (largest deviation: +11.31, i.e. +1.07). The replication extends to the intermediate thresholds (three seeds each at clip 2/3/5; Table S56).

Interpretation and caveats. The readout signature and the probe response are clipping-regime properties of the released recipe, constructible by disabling clipping (or, at intermediate strength, dialled to the released magnitude: clip 5.0 gives readout 82.69/68.4% vs. the released 78.8/70.7%, the closest single match of the ladder). The released operating point is bracketed but not matched in all coordinates—no single threshold places dev 13.38, readout 78.8, REC 12.78, and gap +10.24 simultaneously—so at least one further quantitative difference (CTC normalisation, stopping point,

framework version, or data-order effects) persists; because the released config implies no clipping, the mismatch indicates the released run’s trajectory was slower than ours in dev-competence terms, not that it was clipped. The within-regime readout–gap association of the canonical panel ($\rho=-0.56$; Sup. D) must not be extrapolated across regimes: across the ladder it is strictly positive (readout and gap rise together at every step from clip 1.0 to none).

23 Additional analyses: split reconciliation, dev-split consistency panel, multi-seed ladder, and relocated tables

This appendix collects additional analyses and tables relocated from the main text for length: the ID-level split reconciliation (main §3.1), the dev-split consistency evaluation of the sensitivity checkpoints (main §4.3), multi-seed clip-ladder replicates (main §4.3), and relocated tables.

Table S3: Evidence-graded analysis registry. Rows are grouped by the role each analysis plays in the argument, not by when it was added.

Evidence role	Analysis	Status	Multiplicity handling
<i>Selected (descriptive)</i> <i>probe</i>	PURE-REC gap, original evaluator (post-selection)	D	descriptive MC estimate $p \leq 10^{-4}$; post-selection—see probe multiplicity table in main §4.1
<i>Confirmatory</i>	605-query donor-pool factorial (cells, effects, interaction)	C	Holm across 3 contrasts
	Template-holdout, BT-retrain, cross-fit	C	paired bootstrap CIs
	Protocol checks (scorer, FPS, decoding sweep)	C/D	exact-match / descriptive
	Frozen-protocol confirmation (seeds 1506/1607)	C	confirms family-side pattern
<i>Triangulating</i>	14 reconstruction gaps + prediction interval	C/D	DiD CIs; PI descriptive
(Finding 1: non-replication)	Cross-metric gap table	D	direction/sign only
	Conditioning controls (retrievers, gloss, random)	D	descriptive
	Competence gate, trajectories, Pareto	D	gate counts; 12-threshold grid
	Knowledge-distillation ladder + per- α	D	15 with gap + dev
<i>Triangulating</i>	Reference replacement (single/dual); overlap strata	D	paired bootstrap of gap difference
(Finding 2: reference sensitivity)	Asymmetric floor-effect control	D	BLEU-4/BLEU-1/chrF drops
	Evaluator $\times$ reference (7 $\times$ 3)	D	descriptive decomposition
<i>Triangulating</i>	Five diagnostic views (readout-with-generalization)	D	triangulation, not independent
(Finding 3: readout signature)	In-sample readout (train/dev/test)	D	no identity claim beyond train.pt
	Leakage sanity checks (permutation, equivariance)	D	exact-match / permutation
<i>Robustness</i>	Donor-pool substitution; canonical B/C/D path	D	query-index bootstrap CIs
	Alternative paths, Shapley, S4 system; balance/caliper/signer	D	path-sensitivity display
	Rescue families (20); big-arch; long-schedule	D	competence attempts
	Released-checkpoint perturbation sweep	D	lockstep collapse with competence
	IPW pool-origin robustness	D	balance robustness check
	Rank stability (9 systems $\times$ 7 evaluators)	D	τ_b/ρ /flip means
	Competence extrapolation (OLS/GP)	D	non-IID; LOO CV (5 folds)
	Per-seed extension cells (8 seeds)	D	descriptive; PI only
	Oracle/white-box transfer matrix	D	permutation check
	Template-density strata; donor-exclusion τ -sweeps	D	descriptive
	CSL-Daily readout-ratio check	D	second-dataset boundary
	Transcript-slot diagnostic + extractor audit	D	extractor P/R by system
<i>Administrative</i>	Ethics/licensing declarations	—	updated each revision

Table S4: Canonical checkpoint registry. 76 training runs across 12 families (disk-scanned; auto-generated from `results/canonical_checkpoint_registry.json`). Of these 76 runs, 74 were beam-3 decoded on PHX-public under the canonical §3.2 donor registry. The 74 decoded runs reduce to 73 unique weight binaries: the long-schedule seed-202 artifact is byte-identical to the validation-freq-misread seed-202 artifact (same wandb run, identical per-step losses, identical best-checkpoint SHA-256) and counts once. 69 non-degenerate decoded training runs enter the non-replication count (all 69 with strictly negative gaps; all share the clip=1.0 inference, resolved in Sup. U). Each family row’s columns sum to the Total row.

Family	Runs	Trained	Has gap (beam-3)	Has dev	Enters
Faithful (train_faithful.py, 42–49)	8	8	8	8	RQ1 primary; non-reproduction (clip=1.0 panel; resolved Sup. U)
Reconstructions (primary)	6	6	6	6	multiseed, bootstrap, dose-resp.
Reconstructions (extension)	8	8	8	8	prediction interval, dose-resp.
Validation-freq-misread	4	4	4	4	dose-resp.
Step-corrected (re-trained)	10	10	10	10	recipe-adherence; dose-resp.
Joint-loss (greedy, 1801–1808)	8	8	8	8	protocol-robustness
Joint-loss (beam-3, 1901–1908)	8	8	8	8	protocol-robustness
Joint-loss supplementary (1809/1810)	1	1	0	1	exploratory; no canonical gap
Train-pool ladder	4	4	4	4	dose-resp., gap ranges
Rescue (dropout/wd; only wd0 archived)	1	1	0	1	gate
Confirmation	2	2	2	2	dose-resp., family-side confirm.
Long-schedule	1	1	1	1	gate; best seed dose-resp.
Distillation students	15	15	15	15	ladder, dose-resp.
Total	76	76	74	76	
Released original	—	—	1	1	audit target (not trained)

Per-family columns sum to the Total row. Has-gap counts reflect the disk-scanned registry: 74 beam-3-decoded training runs; one validation-freq-misread seed-202 / long-schedule collision reduces unique binaries to 73. Five of the 74 decoded runs are degenerate (three $\alpha=1.0$ distillation students with empty output and two smallest ladder fractions with BLEU ≈ 0), leaving 69 non-degenerate runs that enter the headline non-replication count. The large-arch (4), rescue-lr (8), rescue-expanded (11 of 12), cross-fit (2), and BT-retrained-holdout (1) families listed in earlier registries were not recoverable from disk after the 2026-07-14 host compromise documented in our security incident report; their family-level summary statistics had been retained in the manual legacy registry but cannot be reproduced from current artifacts, and they are omitted from this disk-scanned table. The 11 released-weight fine-tune variants (Gaussian noise, fine-tuning) live under `checkpoints/finetune.released/` and are NOT counted as from-scratch training.

Table S5: Spearman correlations across the current uniform-readout set (38 constructible plus the released evaluator; 37 constructible have canonical gaps). “Non-degenerate” excludes the three $\alpha=1.0$ distillation students that decode to empty output. Dev-based pairs use the checkpoints that also have uniform dev decodes. Supersedes the earlier 31-point table (`results/readout_gap_correlation.json`).

Pair	ρ	p	n
Readout vs. dev BLEU-4 (incl. released)	0.880	1.5×10^{-6}	18
Readout vs. dev BLEU-4 (excl. released)	0.858	1.1×10^{-5}	17
Readout vs. PURE–REC gap (incl. released)	−0.444	0.0053	38
Readout vs. PURE–REC gap (excl. released)	−0.564	0.00028	37
Readout vs. gap (non-deg., incl. released)	−0.319	0.062	35
Readout vs. gap (non-deg., excl. released)	−0.438	0.0095	34
Gap vs. dev BLEU-4 (incl. released)	0.502	0.040	17
Readout vs. train NLL	−0.561	0.030	15
Readout vs. train token accuracy	0.509	0.052	15

Table S6: Claims register: estimand, unit, selection status, uncertainty, and evidence tier per headline claim (D descriptive, S sensitivity, C prospectively specified within-project, NI not identified).

Claim	Estimand	Unit	Selection status	Uncertainty	Tier
Released probe gap +10.24	PURE–REC sacre-BLEU under released ckpt	item / donor	post-selected max-gap probe cell	donor-cluster bootstrap (OR1 Sup. G)	D
Non-reproduction clip 1.0	at sign of gap per reconstructed run	run (checkpoint)	families post-hoc; 6 legacy seeds prospective threshold grid post-hoc; dev-split consistency panel	seed-dispersion interval, LOO folds	D (C for 6 seeds)
Signature reproduction unclipped	readout, EM, gap of unclipped runs	run (seed)	post-hoc; dev-split consistency panel	3 seeds per threshold, test and dev	S
Reference-frame contrast 9.08	$\text{gap}_{\text{orig}} - \text{gap}_{\text{human}}$, 461 matched items	item	post-hoc subset (confidence=1)	paired bootstrap CI	D
Pool-origin contrasts +14.5/ +9.6/ +8.4	SEEN–UNSEEN gap under three designs	item / resample	post-hoc	bootstrap / conditional simulation	D
Seven-correction competence	dev, test BLEU vs. released	run (seed)	seeds fixed (42–49)	range $\pm$ SD over 8 seeds	D
Readout–gap association	Spearman within/across regimes	ρ checkpoint	post-hoc panel	no CI (rank statistic)	D
Best-seed test 13.11	max over 8 seeds	run	post-hoc maximum	none (reported, not relied on)	D
Update-level diagnostics (§24)	update norms, component norms	optimiser step	single seed (42), lockstep dual run	per-step trajectories (no CI)	S

Table S7: Path sensitivity of the donor-pool substitution decomposition under the original evaluator (common 622-query support; 10,000-resample query-index bootstrap CIs). S1=SEEN-PURE (7,060-pool, max-Jaccard), S2=SEEN-RAND640 (640-pool, max), S3=SEEN-MATCHED (7,060-pool, Jaccard-matched), S4=SEEN-RAND640-MATCHED (640-pool, matched), U=UNSEEN-PURE (test pool). Every ordering telescopes to the same total (A) = S1−U = +14.7.

Ordering	Step	Size contrast	Selection contrast	Origin contrast
Canonical (prespecified)	S1→S2→S3→U	+7.2 [+5.9, +8.4]	−0.8 [−1.8, +0.2]	+8.4 [+7.3, +9.4]
Alternative 1	S1→S2→S4→U	+7.2 [+5.9, +8.4]	+1.8 [+1.1, +2.5]	+5.8 [+4.7, +6.8]
Alternative 2	S1→S3→S4→U	+2.6 [+1.7, +3.5]	+6.4 [+5.1, +7.7]	+5.8 [+4.7, +6.8]
Shapley average	over orderings	+4.9 [+4.0, +5.8]	+4.1 [+3.3, +4.9]	+5.8 [+4.7, +6.8]

Table S8: Factorial design and balance (605 targets). SMD = standardized mean difference, seen vs unseen within stratum.

Quantity	seen_high	seen_low	unseen_high	unseen_low
LaBSE similarity (mean)	0.813	0.654	0.776	0.622
Duration (frames, mean)	100.2	99.5	99.2	98.8
Word count (mean)	13.3	13.6	13.3	13.4
Same-signer rate	100%	100%	100%	100%
Similarity SMD (seen − unseen)	+0.41 (high)		+0.38 (low)	

Table S9: Full 2×2 exposure factorial (605 targets; corpus sacreBLEU in BLEU points; NLL per token). CIs: 10,000-resample bootstrap (original) and seed-level t across six reconstructions; Holm correction across three contrasts.

Metric	Evaluator	seen_hi	seen_lo	unseen_hi	unseen_lo	seen main	sim. main	inter.
BLEU	Original	11.12	2.31	7.19	2.08	+2.08 [+1.6, +2.6]	+6.96 [+5.9, +8.1]	+3.70 [+2.6, +4.8]
BLEU	Seeds (mean)					+0.51	+4.06	−0.39
NLL	Original	3.74	4.30	3.93	4.37	−0.128	−0.498	−0.124
NLL	Seeds (mean)					−0.064	−0.179	−0.004

Table S10: Multivariate donor balance (622-query common support). All $|SMD| < 0.12$.

Covariate	SEEN-MATCHED mean	UNSEEN mean	SMD	Per-query overlap
Source-text Jaccard	0.411	0.415	-0.021	$ \Delta J \leq 0.05$: 96%
LaBSE cosine	0.782	0.774	+0.068	—
Duration (s)	3.76	3.74	+0.014	$ \Delta \leq 1$ s: 57%
Word count	12.29	12.31	-0.003	$ \Delta \leq 2$: 69%
Template density	0.310	0.326	-0.075	—
Same signer as query	0.183	0.228	-0.111	both same-signer: 4.7%

Table S11: Released `config.yaml` load-bearing field-by-field mapping: original-framework semantics (empirically pinned by the released `validations.txt` where noted) vs. our v1–v3 reconstruction semantics (with defects) vs. our seven-correction v4 semantics. Unlisted fields (model architecture, dropout, initialisers, max lengths) are applied verbatim and identical across v1–v4.

Field (value)	(released)	Original semantics	framework	v1–v3 (defect)	semantics	v4	seven-correction semantics
<code>translation_normalization</code> (batch)		batch-sum CE ÷ sentences; logged dev Translation Loss is the raw token sum (43,440.8 = 7,813 × ln 259.84)		CE reduction="mean" (per-token): ~15–30× smaller gradient scale; with Adam weight-decay coupling this changes effective regularization		CE reduction="sum" ÷ batch sentences; smoke-run step-14 dev token-sum 42,981.5 vs. released 43,440.8 (1.1% off)	
dev NLL token count		counts (mask valid)	<i>valid</i> target True =	(<code>txt_mask.sum()</code> counts <i>padding</i> positions (all of v1–v3); also truncates v2 greedy-selection hypotheses/references by padding count		<code>txt_mask.sum()</code> ; and per-token NLL computed on valid tokens only	
<code>recognition_loss_weight</code> (1.0)		CTC+translation; logged Recognition Loss column normalisation not exactly identifiable (17.28 at step 14 matches neither per-sentence 39.9 nor per-gloss-token 12.3 in our stack)		v1: CTC omitted entirely; step-corrected family: <code>--rec-weight 0</code> (CTC off); v2/v3: joint with per-token-mean CTC		joint CTC+translation, CTC token-sum ÷ sentences (batch-consistent); recognition-column normalisation documented as not identifiable	
<code>validation_freq</code> (14)		every 14 <i>optimiser steps</i> (logged Steps: 14, 28, ...)		v1: misread as 14 <i>epochs</i> ; v1.5+/v2/v3: 14 steps (corrected)		14 optimiser steps	
<code>eval_translation_beam_size</code> (3) for selection		beam-size; progressive decode for validation BLEU (WER/BLEU columns of released log)	beam-3	v1: selection branch never saved (hard-coded <code>improved=False</code>); v1.5/v2: teacher-forced greedy argmax, not autoregressive; v3: beam-3 (corrected)		autoregressive beam-3 (via <code>back_translate</code>), identical to the evaluation decoder	
<code>epochs</code> (3000) / stopping		released run ended at step 2828 (101 epochs) after 72 consecutive non-improving validations—patience-5 early stop is <i>provably not</i> the released rule; LR-floor is the only config-reproducible stop		300-epoch cap with early-stop patience 15		3000-epoch cap; default stop rule = LR floor (1e-8); patience rule available but off and documented as not release-matching	
<code>scheduling</code> (plateau), <code>decrease_factor</code> (0.8), <code>patience</code> (5)		ReduceLROnPlateau(mod <i>present</i>) since v1, but on dev BLEU; every logged decay fires on the 6th consecutive non-improving validation		reached only ~60–90 validations before the patience-15 stop		identical scheduler wiring; LR logged after <code>scheduler.step</code> as in the released log	
<code>batch_size</code> (256)		256 sentences per optimiser step		v1: physical 64 × grad-accum 4 (inferred); v1.5+/v2/v3: 256		256	
<code>random_seed</code> (42)		42 (released run)	30	never used as the primary seed (101–1405, 1701–1908 sweeps)		42 is the primary run; 43–49 as within-implementation dispersion	
gradient clipping		not specified in config ⇒ signjoey framework default no clipping (build <code>gradient_clipper</code> returns <code>None</code> when <code>clip_grad_val/clip_grad_norm</code> are both absent; the framework implements <i>both</i> value clipping and norm clipping, selected by the respective field)—resolved as the eighth defect (Sup. U)		hard-coded <code>max_norm=1.0</code> (with token-mean loss it almost never binds)		<code>--clip 1.0</code> default, disclosed as author inference; with batch-normalised loss the norm genuinely binds (first steps 377/568/145)— <i>overturned</i> : unclipped reproduces the signature and probe response (Sup. U)	

Fields applied verbatim in all our versions (not shown): `learning_rate` 0.001, `betas` [0.9, 0.998], `weight_decay` 0.001, `label_smoothing` 0.0, `learning_rate_min` 1e-8, `keep_last_ckpts` 1, architecture block (transformer, 256/512/8/3, dropout 0.1, softmax, xavier), `eval_translation_beam_alpha` -1, `translation_max_output_length` 30

Table S12: Epoch-validation approximation (validate every 14 epochs, not steps): 300 epochs, dev-NLL selection. All four seeds produce negative PURE-REC gaps under the canonical §3.2 donor registry; none shows the reversal or readout signature.

Seed	ep.	dev NLL	REC	PURE gap
101	27	2.11	10.12	−0.60
202	—	—	10.14	−1.70
303	—	—	8.74	−1.51
404	—	—	8.43	−0.52
Released evaluator	1,820	—	12.78	+10.24

Table S13: Pose-perturbation battery (corpus sacreBLEU, BLEU points) under the released evaluator and the best rescue checkpoint (wd0, seed 202). The TN-PURE-v1 identity column here (23.79) reflects the pre-exclusion materialization used at the time this battery was decoded; the canonical materialization gives 23.02 (gap +10.24). Because the battery varies the pose and re-decodes, not the donor choice, the perturbation differentials are unaffected by the materialization difference (Sup. N: mean $|\Delta\text{gap}| = 0.65$, max 1.45).

Evaluator	System	identity	reversal	window shuffle	jitter 5%	hand mask	face mask
Original	REC-v1	12.78	7.25	11.97	12.51	1.35	3.15
Original	TN-PURE-v1	23.79	7.72	18.35	23.91	1.30	4.06
wd0 (seed 202)	REC-v1	10.73	6.26	10.54	10.60	0.63	1.58
wd0 (seed 202)	TN-PURE-v1	10.98	6.66	9.65	10.92	0.58	1.43

Table S14: Cluster-aware bootstrap for the headline PURE-REC gap (canonical §3.2 registry, original refs, released-evaluator hypotheses from `results/gap_43_canonical_beam3_items/released_{gt,pure}.json`). All CIs are 10,000-resample sacreBLEU `tok:13a` exp-smooth over precomputed per-item n-gram sufficient statistics. With so few clusters (especially show with only 2), these are *exploratory sensitivity checks*, not confirmatory tests. Donor reuse: 641 queries select 565 unique donors (61 donors serve more than one query; maximum reuse 6; Gini 0.11).

Analysis	Design	Clusters	Gap	95% CI
Query-level (baseline)	item	641	+10.24	[+8.88, +11.62]
Donor cluster	donor	565	+10.25	[+8.87, +11.64]
Donor two-way (query×donor)	nested	565	+10.25	[+8.79, +11.74]
Signer	signer	9	+10.30	[+9.29, +11.39]
Broadcast show	show	2	+9.27	[+5.16, +11.44]
Broadcast date	date	297	+10.25	[+8.83, +11.72]
Signer×Show	combined	17	+10.14	[+7.89, +11.91]

Table S15: PHX-public pure-donor source-text donor-exclusion τ -sweep under the original SLRTP2025 BT (ical tie-breaking, 12.5 fps; sacreBLEU on the 0–100 scale). The retrieval-exceeds-REC reversal survives exact exclusion and $\tau=0.5$ exclusion.

Condition	sacreBLEU	Δ vs REC	Rev
REC baseline	12.8	—	
Text-nearest pure (no exclusion; pre-exclusion historical materialization, Sup. N)	23.8	+11.0	
Text-nearest pure (canonical, w/ exclusion)	23.02	+10.24	
Exact-text excluded	23.0	+10.2	
$\tau=0.7$ (Jaccard > 0.7 excluded)	21.2	+8.4	
$\tau=0.5$ (Jaccard > 0.5 excluded)	17.8	+5.0	
$\tau=0.3$ (Jaccard > 0.3 excluded)	8.0	−4.8	
$\tau=0.1$ (Jaccard > 0.1 excluded)	0.2	−12.5	
Random donor (mean of 20 seeds)	0.9	−11.6	

Table S16: Template-density stratification (sacreBLEU 0–100). Retrieval exceeds REC at all density levels.

Density stratum	Mean density	REC	Retrieval	Gap
Low (n=249)	0.152	5.9	16.6	+10.7
Mid (n=179)	0.254	9.5	19.7	+10.1
High (n=213)	0.510	26.1	38.6	+12.4

Table S17: Holdout boundary controls (sacreBLEU, 0–100; historical—per-item artifacts retained, training runs unrecoverable after the 2026-07-14 host incident). None reproduces the canonical +10.24 reversal.

Control	Setup	REC	Retrieval	Gap
Same evaluator	test split (template-overlap)	12.8	23.02	+10.24
Same evaluator	template-holdout (seen in training)	77.0	20.1	−56.9
BT-retrained (unseen holdout)	train-core 6,535	6.2	5.4	−0.9
Retrained BT on PHX-public	unseen to both	8.7	8.3	−0.4
Cross-fit A/B (donor-unseen)	pooled, $n=659$	4.0	3.9	−0.1

Table S18: Frozen template-holdout controls (1,538 windows, $\alpha=0.5$; sacreBLEU 0–100).

System	sacreBLEU	Empty %	Mean Jac.
REC baseline	13.10	9.95	—
Random train donor	0.04	11.83	—
Text-nearest train donor	2.30	10.66	—
LaBSE / multi-SBERT (donor copy)	1.73 / 1.51	9.2 / 11.3	—
Noisy-gloss retrieval (strong NMT)	0.40	10.90	—
Retrieval composition ($\alpha=0.5$)	2.20	10.79	—
$\tau=0.1$ / 0.3 / 0.5 / 0.7 exclusion	0.04 / 1.05 / 1.93 / 2.01	10.0–10.5	0.10/0.29/0.40/0.42
BM25 / TF-IDF nearest	1.75 / 2.36	10.8 / 8.8	—

Table S19: Competence extrapolation: gap predictions at dev BLEU-4 = 13.38 from three fits over 65 unique non-released binaries (dev BLEU-4 0–10.5; 61 non-degenerate). All predictions are far below the observed canonical +10.24. PIs are *descriptive* (non-IID checkpoints); they should not be read as calibrated uncertainty intervals.

Fit	R ² / LML	Prediction at 13.38	95% PI (descriptive)
OLS linear	0.48	−1.84	[−2.67, −1.01]
OLS quadratic	0.50	−1.09	[−1.83, −0.35]
Gaussian Process (RBF + white)	−48.1	−1.00	[−2.16, +0.17]
Released evaluator (observed, canonical)	—	+10.24	[+8.88, +11.62]

Table S20: Leave-one-family-out cross-validation of the competence extrapolation. Each row refits all three models after removing one checkpoint family (ladder fractions are grouped with the distillation family, matching the §4.2 figure family mapping). The GP upper bound stays below +0.3 across all folds; OLS quadratic is the most sensitive (shifts sign when distillation is removed). All intervals descriptive.

Family removed	n remaining	OLS-lin pred	OLS-quad pred	GP pred	GP upper	OLS-lin R ²
None (full)	42	−1.84	−1.09	−1.00	+0.17	0.48
Reconstructions (14)	28	−1.84	−2.07	−0.89	+0.25	0.57
Distillation (19)	23	−0.78	+0.23	−1.18	−0.41	0.03
Validation-freq-misread (8)	34	−1.89	−1.32	−0.97	+0.22	0.56
Rescue (1)	41	−1.82	−0.97	−0.98	+0.18	0.47

Table S21: CSL-Daily Mode B donor-exclusion (1,176 sequences; 0–100 scale).

Condition	sacreBLEU	BLEU-1
REC baseline	26.7	79.1
Text-nearest donor	1.6	43.5
Random donor	0.0	3.9
$\tau=0.5$ exclusion	1.0	42.6
BM25 retrieval	1.5	39.2
TF-IDF retrieval	1.2	38.5

Table S22: Label permutation test (1,000 items, reference labels shuffled *after* decoding). BLEU collapses from 78.6 to ~ 1.0 , confirming output–reference alignment. This test alone cannot distinguish pose-conditioned output from ID/cache-based retrieval; the input-side pose permutation below addresses that.

Condition	BLEU	EM
Original (correct pairing)	78.61	0.701
Permutation 0	1.06	0.000
Permutation 1	1.30	0.000
Permutation 2	0.90	0.002

Table S23: Input-side pose permutation with output-equivariance (100 items). Pose tensors are shuffled *before* decoding while IDs stay fixed. The equivariance column verifies $E(\text{pose}_j, \text{slot}_i) = E(\text{pose}_j, \text{slot}_j)$.

Condition	BLEU vs slot ref	Detail
Original decode vs own reference	75.62	baseline
Shuffled poses vs slot reference	0.95	BLEU collapses
Shuffled poses vs donor reference	75.62	identical to original
Output-equivariance match rate	—	100/100 (100%)
Donor-length match rate	—	100/100 (100%)
Slot-length match rate	—	5/100 (5%)

Table S24: Stratified exact-match rate on the full 7,060-item training pool under beam=3 free-decode.

Stratum	n	EM
By signer		
Signer01	1,852	0.683
Signer02	83	0.590
Signer03	579	0.736
Signer04	1,049	0.748
Signer05	1,623	0.701
Signer06	35	0.714
Signer07	762	0.696
Signer08	844	0.715
Signer09	233	0.734
By reference length		
Short (≤ 5 words)	203	0.837
Medium (6–10)	1,319	0.685
Long (11–20)	4,335	0.714
Very long (> 20)	1,203	0.686
By text repetition in train		
Unique (1 occurrence)	6,777	0.702
Low (2–3)	57	0.737
Mid (4–10)	76	0.684
High (> 10)	150	0.940

Table S25: Same-signer held-out control: train (seen) vs. test (unseen) free-decode. The 67-point BLEU gap is consistent with training-pool exposure contributing to the readout, though the lack of full covariate matching means we cannot quantify how much of the gap is exposure vs. distributional shift.

Split	n	BLEU	EM	Shared signers
Train (seen)	7,060	78.82	0.707	9
Test (unseen)	500	12.19	0.032	9 (same)

Table S26: Three-way data provenance for PHX-public. The released evaluator’s training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs carry the missingness label `absent_from_slrtp2025_released_pose_records`—a status flag rather than an explained mechanism (SI Sup. A). SHA-256 hashes verify file integrity.

Split source	Train	Dev	Test
Official PHOENIX-2014T	7,096	519	642
Released SLRTP2025 pose (.pt)	7,060	515	641
Released evaluator <code>config.yaml</code> manifest	7,060	515	641
Missing from official	36	4	1

Table S27: Checkpoint accounting, auto-generated from `results/accounting-table.json` (single source of truth, disk-scanned). “Runs” = from-scratch training executions; “Decoded” = beam-3 decoded on the canonical PHX-public gap panel; “Unique” = distinct SHA-256 weight binaries; “Non-deg.” = non-degenerate (BLEU > 0, excluding 3 $\alpha=1.0$ distillation students with empty output and 2 smallest ladder fractions). The partially-corrected tier (14 corrected-recipe runs) provides recipe-adherence evidence; secondary (14 legacy, of which 6 prospectively specified) provides the prospectively specified non-reproduction direction plus stability; diagnostic is mechanism exploration.

Tier / Family	Runs	Decoded	Unique	Non-deg.
Faithful	8	8	8	8
Faithful (train_faithful.py, seeds 42–49)	8	8	8	8
Partially-corrected	14	14	14	14
Validation-freq-misread	4	4	4	4
Step-corrected (re-trained)	10	10	10	10
Secondary	14	14	14	14
Paper-derived (prospectively specified)	6	6	6	6
Paper-derived (extension seeds)	8	8	8	8
Diagnostic	40	38	37*	33
Joint-loss (greedy selection)	8	8	8	8
Joint-loss (beam-3 selection)	8	8	8	8
Joint-loss (supplementary, 1809/1810)	1	0	0	0
Confirmation	2	2	2	2
Rescue	1	0	0	0
(dropout/wd variants)				
Distillation	15	15	15	12
Ladder (data fractions)	4	4	4	2
Long-schedule	1	1	0*	1
Total trained	76	74	73	69

*One validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact (same wandb run); it counts once in the unique-binary column (total unique = 74 decoded – 1 collision = 73). The released evaluator (1 checkpoint) is not a training run and is not included above. Eleven released-weight fine-tune variants (Gaussian noise, local perturbation) are not from-scratch training and are not counted here. The previously-reported large-arch, rescue-lr, cross-fit, and B37-retrained holdout families were not recoverable from disk after a host compromise (SI Sup. D) and are omitted; their family-level summary statistics had been retained in the legacy manual registry but cannot be reproduced from artifacts.

Table S28: Novelty relative to closely related PHOENIX-2014T and learned-metric reproducibility studies.

Study	Primary finding	Method	Distinction from present work
Czehmann et al. (2026)	Reference replace-ment changes BLEU scores and rankings on PHOENIX-2014T	Human back-translations of test set	We apply the matched-subset protocol to the released evaluator with a frozen replay probe and quantify the paired reference-frame contrast
Alkain et al. (2026)	Highest train-test text overlap among tested SLT corpora	N-gram overlap quantification	We use their finding to contextualize why PHX-public is particularly susceptible to reference coupling
Artiaga et al. (2025)	Signer-dependent evaluation overestimates SLT capability	Signer-disjoint splits	We include signer as a regression covariate (not signer-disjoint evaluation; ΔR^2 from removing the signer block is $\leq 1\%$)
Jiang et al. (2025)	BT likelihood is a more stable correlate of human judgment than decoded BLEU	Meta-evaluation of pose/embedding/BT metrics	We report that teacher-forced likelihood does not reproduce the reversal (REC NLL 3.060 vs PURE 3.479)
Chen et al. (2022)	BERT-based metrics fail to reproduce due to missing preprocessing/code	Multi-implementation replication	We extend the learned-metric reproducibility question to a full SLP evaluator pipeline with 68 training runs across 11 families
This work	Our disclosed reconstruction from the public artifacts did not attain the released evaluator’s competence or replay-probe behavior	68 training runs across 11 families, fixed-output cross-evaluator re-decoding, reference/donor substitution	—

Table S29: Public-artifact provenance of each reconstruction family. “Dev BLEU-4” is the uniform full-pool beam-3 value (§3.4); the released evaluator scores 13.38. “Selection” is the checkpoint-selection rule. The seven-correction runs implement the released config exactly; the paper-derived runs use NLL selection on a legacy script; the step-corrected runs match the released `validation_freq=14` JoeyNMT semantics with decoded-BLEU selection; the joint-loss variants additionally correct the recognition-loss omission.

Family	Public source used	Known deviation	Runs	Selection	Dev BLEU-4
Faithful	released config.yaml exactly	none (seven defects corrected; seed 42 first)	8	beam-3-BLEU	11.59–12.62
Paper-derived (prosp.-spec. + ext.)	config.yaml + architecture	legacy script; epochs; NLL selection	300 14	NLL on dev	6.5–9.4
Validation-freq-misread	config.yaml (decoded-BLEU)	<code>validation_freq=14</code> misread as epochs initially	4	decoded BLEU	8.3–9.8
Step-corrected (re-trained)	config.yaml JoeyNMT semantics	+ fixed step interpretation; translation-only loss	<code>validation_freq</code> 10	greedy-BLEU	7.6–10.5
Joint-loss (greedy)	config.yaml (<code>recognition_loss_weight=0</code>)	adds CTC; token-mean selection	8	greedy-BLEU	6.3–10.2
Joint-loss (beam-3)	config.yaml (<code>eval_beam=3</code>)	adds CTC; token-mean loss; beam-3 selection	8	beam-3-BLEU	8.7–10.5
Confirmation	recipe-adherent protocol	independent re-run for reproducibility	2	decoded BLEU	9.0–9.5
Long-schedule	config.yaml (epochs=3000)	extended training budget	1	decoded BLEU	≤ 10.02
Rescue (dropout/wd)	hyperparameter grid	dropout/wd variations; none passed gate	1	decoded BLEU	≤ 10.02
Distillation	released checkpoint as teacher	student-teacher KL; students trained from scratch	KL; 15	decoded BLEU	7.0–9.0
Ladder (data fractions)	config.yaml	data-subset fractions (0.125–0.75 $\times$)	4	decoded BLEU	≤ 9.8
Joint-loss (supplementary)	exploratory v2 seeds (1809/1810)	out-of-protocol v2 explorations	1	—	not decoded
Total			76		

Table S30: Teacher-forced and free-decode diagnostic statistics for the released evaluator and fourteen reconstruction seeds (full table in `results/checkpoint_stats.json`). Teacher-forced values on 500 randomly sampled items (seed 42, sampled without replacement; the sampling SD across 5 re-samples is ≤ 0.3 pp for token accuracy and ≤ 0.01 for NLL/tok); free-decode values are from the training pool (7,060 train / 515 dev) under the uniform beam-3 protocol (§3.4). sacreBLEU signature: `sacreBLEU|nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.5.1` (max n-gram order 4, the default).

Checkpoint	Train tok. acc.	Train NLL/tok	Train free-decode BLEU	Dev NLL/tok	Dev free-decode BLEU
Released	0.965	0.158	78.82	3.236	13.8
Reconstructions (range)	0.550–0.596	1.72–1.92	7.20–9.25	2.63–2.68	6.49–9.25
Reconstructions (mean)	0.572	1.82	8.35	2.66	8.35

Table S31: Custom-terminology registry. Each term maps to its estimand, the dataset on which it is computed, and its evidence grade (all descriptive).

Term	Estimand	Dataset	Grade
Training-pool read-out	In-sample autoregressive free-decode BLEU/EM	Full 7,060-item training set	D
PURE-REC gap	Corpus sacreBLEU-4 difference (PURE – REC)	641-item PHX-public test	D (post-selection)
Operational tolerance gate	One-sided: dev BLEU-4 ≥ 12.38 and dev WER ≤ 86.37	515-item dev set	D

Table S32: Claim hierarchy. Each key claim mapped to its evidence tier and supporting evidence.

Claim	Tier	Evidence
Our disclosed reconstruction did not attain released evaluator competence	D	28 recipe-conditioned runs, all dev BLEU < 10.5; not a public-artifact-insufficiency claim
Probe response does not reproduce across canonical (clip=1.0) constructible checkpoints	D	69/69 non-degenerate gaps negative; resolved by the clipping sensitivity (Sup. U: unclipped +11.19)
Probe response reflects a causal checkpoint-identity effect	NI	Resolved as a clipping-regime property; no identity effect remains to attribute
Headline gap +10.24 is post-selection	S	Max-gap probe among 8 explored; CI descriptive
Reference-frame contrast: gap falls 90.5% under human back-translations	S	Single-annotator, 461-item, 4 confounded changes
Origin contrast persists under matched pools (+9.56)	S	148 items, SMD<0.1; association not causal
Probe is adversarial lexical-coupling stress test	D	Source = reference in PHX-public; retriever uses source Jaccard

Table S33: Decoded-text metric sensitivity under the original SLRTP2025 evaluator (641 sequences). The reversal direction is consistent across decoded-text metrics; teacher-forced NLL does not reproduce it (SI Sup. O). Item-level pose-DTWp is not reported (see main text).

System	BLEU-4	BLEU-1	chrF	ROUGE-L	WER	BERTSc.
REC-v1 (recorded test poses)	12.78	34.40	34.59	35.20	85.77	0.757
PT-v1	1.59	16.91	19.03	17.26	101.68	0.669
TN-PTCOMP	18.86	46.35	41.88	45.17	77.36	0.793
TN-PURE	23.02	55.43	47.82	53.07	71.87	0.822
Gap PURE-REC	+10.24	+20.94	+12.93	+17.87	-13.90	+0.066

Table S34: Asymmetric floor-effect control (full-641 sensitivity analysis). PURE loses a larger fraction of its REC across all metrics when references are replaced, so simple symmetric compression is insufficient; the model does not discriminate between reference-text defects and training-distribution alignment.

Metric	REC (orig)	REC (human)	REC drop	PURE (orig)	PURE (human)	PURE drop
BLEU-4	12.78	5.66	7.11 (55.7%)	23.02	6.52	16.50 (71.7%)
BLEU-1	34.40	29.28	5.12 (14.9%)	55.43	32.35	23.08 (41.6%)
chrF	34.59	32.75	1.84 (5.3%)	47.82	35.02	12.80 (26.8%)
Dual-ref gap (orig+human)	BLEU-4: +11.50; BLEU-1: +20.03					

Table S35: PURE-REC gap across metrics and evaluators on PHX-public (641 sequences; canonical decoding) for the original evaluator and the *six primary* reconstructions (of fourteen; the eight extension seeds show the same near-zero/negative pattern in the main-text multiseed factorial table). chrF, WER and ROUGE-L use the official SLRTP2025 implementations. WER and NLL are sign-flipped so positive always means PURE is better than REC. BERTScore-F1 uses `bert-base-multilingual-cased` (bert-score 0.3.13 package, whose module `__version__` string reads 0.3.12; `lang=de`, `idf=False`, no baseline rescaling; model revision logged in the artifact); per-checkpoint BERTScore cells are in the artifact. The large reversal is consistent in direction across decoded-text metrics under the original evaluator and disappears under all six primary reconstructions; residual reconstruction signs are mixed and near zero for n-gram metrics.

Evaluator	Δ BLEU-4	Δ BLEU-1	Δ chrF	Δ WER [†]	Δ NLL [†]	Δ BERTSc.
Original	+10.24	+20.94	+12.93	+13.90	-0.419	+0.066
Seed 101	-0.80	-1.23	-1.44	-2.95	-0.219	-0.005
Seed 202	-1.30	-1.33	-1.57	-3.52	-0.233	-0.006
Seed 303	-1.85	-1.99	-2.26	-1.53	-0.240	+0.001
Seed 404	-1.81	-1.60	-1.90	-1.87	-0.194	-0.001
Seed 505	-1.28	-1.53	-1.59	-3.41	-0.268	-0.004
Seed 606	-1.24	-1.26	-1.55	-3.46	-0.169	-0.001

[†]WER and NLL are sign-flipped: positive means PURE is better than REC (lower WER, lower NLL).

Table S36: Public SLRTP2025 challenge leaderboard on ph14t-test (Walsh et al. Walsh et al. (2025), Table 1), computed under the same released BT evaluator the main text audits; the “Ground Truth” row matches the canonical REC exactly. USTC-MoE is roughly even with REC on lexical BT metrics (BLEU-4 -0.72 , BLEU-1 $+0.45$, chrF $+2.24$, WER $+7.72$) despite a much larger DTW-MJE (0.0448) and duration ratio (1.438). This single-evaluator co-occurrence is descriptive and does not establish evaluator retrieval-favoring bias.

System	BLEU-1	BLEU-4	chrF	ROUGE	WER↓	DTW-MJE↓
Recorded test poses (REC)	34.40	12.78	34.59	35.20	85.77	0.0000
Team 1 – USTC-MoE (retrieval)	34.85	12.06	36.83	36.59	93.49	0.0448
Team 3 – Hacettepe	30.44	9.59	29.70	30.64	88.88	0.0423
Progressive Transformer	22.17	5.43	24.13	21.98	101.45	0.0418
Team 2 – hfut-lmc	16.96	2.05	25.88	19.77	147.85	0.0403
TN-PURE-v1 (replay probe)	55.43	23.02	47.82	53.07	71.87	0.00678

Table S37: Per- α distillation ladder summary. 15/15 students evaluated with the canonical donor registry; “dev BLEU-4 (max)” is the training-time dev BLEU-4 recorded by the original training protocol (legacy validation-selected protocol; not comparable to the uniform full-pool free-decode dev BLEU-4 reported in main §4.2; checkpoints were selected by training-time dev NLL, Sup. E). $\alpha=0$ = hard-label only; $\alpha=1$ = full soft-label KL. All $\alpha<1.0$ yield negative gaps; $\alpha=1.0$ produces empty/degenerate outputs (641/641 or 483/641 empty hypotheses). Train-pool readout uses the identical full-7,060, beam-3, $[::2]$ -subsample protocol for every student and for the released evaluator (no 1,000-item-prefix mixing); the three $\alpha=1.0$ students decode to empty or near-empty strings on both dev and test (uniform full-pool: 515/515 or 390/515 dev hypotheses empty, 641/641 or 483/641 test hypotheses empty) and score 0.00/0.0%; their dev BLEU-4 is degenerate and not reported.

α	n	dev BLEU-4 (max)	gap (range)	train readout (BLEU / EM%)	reaches 10.5?
0.0 (hard-label)	3	10.51	$[-1.71, -0.80]$	8.45–8.87 / 1.9%	1/3
0.25	3	10.75	$[-2.01, -0.77]$	7.81–8.63 / 1.5–1.9%	2/3
0.5	3	10.41	$[-1.38, -0.77]$	7.73–8.20 / 1.7–1.9%	0/3
0.75	3	10.93	$[-1.21, -0.81]$	6.59–7.01 / 1.3–1.4%	2/3
1.0 (full KL)	3	<i>degenerate</i>	$[0.00, 0.00]$	0.00 / 0.0% (empty)	—
Released (full 7,060, beam=3)	1	13.38	+10.24	78.82 / 70.7%	—

Table S38: Canonical checkpoint registry (summary). 76 training runs across 12 families; 69 non-degenerate checkpoints enter the non-reproduction count (all 69 with strictly negative gaps; all share the clip=1.0 inference, resolved in Sup. U), 74 have a canonical PHX-public PURE-REC gap, and the released evaluator is the audit target. Auto-generated from the disk-scanned `results/canonical_checkpoint_registry.json`; full per-checkpoint details in Sup. D.

Family	Runs	Trained	Has gap	Has dev	Enters
Reconstructions (paper-derived, 14)	14	14	14	14	multiseed, bootstrap, PI, dose-resp.
Partially-corrected (config-adherent + step-corrected, 14)	14	14	14	14	recipe-adherence evidence
Joint-loss (greedy + beam-3, 16)	16	16	16	16	protocol-robustness
Other diagnostic (confirmation + rescue + ladder + long-schedule, 8)	8	8	6	8	dose-resp.
Distillation (15)	15	15	15	15	ladder, dose-resp.
Joint-loss supplementary (1809/1810, 1)	1	1	0	1	(exploratory, not decoded)
Total	68	68	65	68	
Released original	—	—	1	1	audit target

Table S39: Experimental protocol summary (moved from the main text).

Setting	Recognizer	Layout	Split size	N eval	α
PHX-public	SLRTP2025 evaluator	178-joint	641 seq	641	0.88 (tuned)
PHX-holdout-MSKA	mska_slr (phoenix-2014t)	133-joint	1,538 win	1,538	0.50 (frozen)
CSL-Daily	mska (csl-daily)	133-joint	256 of 1,176 win	256	0.50 (frozen)

Table S40: Competence rescue sweep: 20 runs across six hyperparameter families all fail the competence gate. Best variant: weight-decay 0, seed 202 (gate threshold defined in the main text).

Variant family	dev BLEU-4 range	min $ \Delta\text{BLEU-4} $ vs Original
lr 5×10^{-4} / 2×10^{-3} (8 runs)	7.9–9.7	3.7
batch size 128 / 512	6.8–9.2	4.2
dropout 0.2	6.9–7.5	5.9
weight decay 0	9.8–9.9	3.5
label smoothing 0.1	6.8–7.4	6.0
long schedule (600 epochs, lr 5×10^{-4})	8.2–8.3	5.1

Table S41: Donor-origin contrast estimands (sacreBLEU, BLEU points; gaps vs local REC). SEEN = 7,060 training poses; UNSEEN = 640 other test poses. Per-seed cells in the artifact.

Evaluator	SEEN-PURE	UNSEEN-PURE	SEEN-REC	UNSEEN-REC	Estimand
Original	23.02	8.51	+10.24	-4.27	+14.51
Seeds (range)	6.83–9.09	6.59–7.98	-1.85 to -0.80	-1.94 to -0.27	-0.15 to +1.22

Table S42: Matched-subset reference sensitivity (corpus sacreBLEU, BLEU points) under the released evaluator. Primary analysis: confidence=1 subset (461 items) with original and human references computed on the *same* items. Sensitivity analysis: full 641 items. Human references: Czehmann et al. Czehmann et al. (2026) back-translations (CC BY-NC-SA 4.0). Confidence breakdown: 461 confidence=1.0, 121 confidence=0.5, 59 confidence=0.0.

Reference condition	<i>N</i>	REC	PURE	Gap
<i>Primary: matched confidence=1 subset</i>				
Original (speech-transcribed)	461	14.94	24.98	+10.03
Human back-translation (conf=1)	461	7.35	8.30	+0.95
Paired reference-frame contrast	9.08 BLEU pts (90.5% gap reduction), CI [7.44, 10.86]; ratio CI [0.81, 1.01]			
<i>Sensitivity: full 641 items</i>				
Original (speech-transcribed)	641	12.78	23.02	+10.24 [+8.88, +11.62]
Human back-translation (full)	641	5.66	6.52	+0.86 [+0.38, +1.91]
Paired contrast (sensitivity)	9.38 BLEU pts (91.6%)			
Human refs, reconstructions (6 seeds)	641	3.49–4.62	3.35–4.35	−0.15 to −0.65

Table S43: Cross-evaluator re-decoding of the noisy-gloss retrieval outputs (641 PHX-public keys; §4.5). Gap = noisy-gloss – REC, corpus sacreBLEU-4 (flat-reference form, §3 protocol). Rows sum to 70 cross-evaluation decodes over 69 unique evaluators (the byte-identical `cf_202/ls_202` pair counted once; the panel is the 74-run gap-decode set plus the rescue run, which was never gap-decoded, so the unique count is one higher than the gap panel’s 68). The realistic retrieval pipeline is roughly even with REC under the released evaluator (−0.13), and every non-degenerate evaluator gives a negative gap; the five degenerate checkpoints (three $\alpha=1.0$ distillation students, two smallest ladder fractions) decode empty or near-empty hypotheses and are excluded.

Evaluator	n	Gap range	Mean gap
Released	1	−0.13 (NG 12.65, REC 12.78)	−0.13
Faithful (seeds 42–49)	8	[−3.22, −1.59]	−2.22
Reconstructions	14	[−1.58, −0.37]	−1.14
Validation-freq-misread	4	[−1.79, −1.00]	−1.51
Step-corrected (re-trained)	10	[−1.36, −0.42]	−0.85
Joint-loss (greedy)	8	[−1.99, −0.84]	−1.43
Joint-loss (beam-3)	8	[−2.17, −0.48]	−1.64
Confirmation	2	[−0.95, −0.67]	−0.81
Long-schedule	1	[−1.79]	−1.79
Distillation students ($\alpha<1.0$)	12	[−1.94, −0.60]	−1.23
Rescue wd0	1	−1.49	−1.49
Ladder (non-degenerate fractions)	2	[−0.58, −0.07]	−0.33

Table S44: Cross-evaluator re-decoding of the rebuilt USTC-MoE fixed outputs (497 of 641 PHX-public keys; §4.5). Gap = USTC-MoE – REC, corpus sacreBLEU-4 (REC recomputed on the same 497 keys, hence 12.36 rather than the 641-key 12.78). The rebuild decodes far below the officially reported leaderboard score (2.42 vs. 12.06), so it is pipeline-equivalent but not score-equivalent to the archived submission; every non-degenerate evaluator gives a negative gap. The eight seven-correction checkpoints were cross-evaluated for the noisy-gloss system (Table S43) but not for this rebuild. The five degenerate checkpoints (three $\alpha=1.0$ distillation students, two smallest ladder fractions) decode empty or near-empty hypotheses and are excluded.

Evaluator	n	Gap range	Mean gap
Released	1	−9.94 (USTC-MoE 2.42, REC 12.36)	−9.94
Reconstructions	14	[−8.30, −5.91]	−7.45
Validation-freq-misread	4	[−7.77, −6.72]	−7.27
Step-corrected (re-trained)	10	[−8.29, −7.01]	−7.67
Joint-loss (greedy)	8	[−9.52, −6.57]	−8.40
Joint-loss (beam-3)	8	[−9.47, −7.98]	−8.91
Confirmation	2	[−6.91, −6.65]	−6.78
Long-schedule	1	[−7.61]	−7.61
Distillation students ($\alpha<1.0$)	12	[−8.26, −6.13]	−7.16
Rescue wd0	1	−7.58	−7.58
Ladder (non-degenerate fractions)	2	[−5.47, −4.28]	−4.88

Table S45: Four-layer reproduction of the headline results from the artifact (§3). Layer 1 regenerates every paper number from committed per-item decodes; Layer 2 re-executes analyses from per-checkpoint data; Layer 3 re-decodes fixed outputs from checkpoint weights; Layer 4 retrains from scratch. **make core-audit** executes L1 only and is not an end-to-end reproduction from training data.

Layer	Command	Scope	Does NOT do
L1: number recomputation	<code>make core-audit</code>	Regenerates canonical donor registry and all headline numbers from committed per-item decodes (asserts REC 12.78 / PURE 23.02 / gap +10.24 / CI [8.88, 11.62])	Does not re-decode or retrain
L2: analysis re-execution	<code>make round2-all;</code> individual <code>make</code> targets (donor-bootstrap, ref-frame-figure, experiment-C/A1/D, full-readout)	Re-runs every analysis from committed per-checkpoint data and hypotheses	Does not re-decode
L3: model re-evaluation	<code>scripts/decode_cells.py;</code> <code>scripts/e_full_readout.py</code>	Re-decodes fixed outputs from checkpoint weights	Does not retrain; requires checkpoint weights + data bundle
L4: from-scratch retraining	<code>scripts/train_reconstruction.py;</code> <code>launch_reconstructions.sh;</code> rescue/distillation/validation-freq-misread launchers	Retrains and re-evaluates from the public recipe	Requires PHOENIX-2014T licence, exact env, ~200 GPU-hours

Table S46: Per-checkpoint teacher-forced and free-decode diagnostic statistics. Teacher-forced token accuracy is the fraction of non-pad tokens where arg max of the teacher-forced logits matches the target; free-decode BLEU uses beam-3 autoregressive decoding on the full training pool (7,060) and full dev set (515) under the uniform protocol. Full data in `results/checkpoint_stats.json`.

Checkpoint	Train tok. acc.	Train NLL/tok	Train free-decode BLEU	Dev NLL/tok	Dev free-decode BLEU
Released	0.965	0.158	78.82	3.236	13.38
Seed 101	0.570	1.822	8.54	2.648	8.51
Seed 202	0.550	1.887	7.62	2.675	6.88
Seed 303	0.596	1.723	8.24	2.672	7.72
Seed 404	0.550	1.922	8.06	2.645	7.33
Seed 505	0.564	1.870	8.58	2.655	9.04
Seed 606	0.574	1.816	7.20	2.672	6.49
Seed 707	0.590	1.718	9.25	2.644	9.37
Seed 808	0.574	1.806	9.11	2.654	8.97
Seed 909	0.565	1.851	8.58	2.654	7.84
Seed 1001	0.584	1.767	8.63	2.682	9.33
Seed 1102	0.584	1.759	8.94	2.632	8.67
Seed 1203	0.574	1.793	8.38	2.659	7.45
Seed 1304	0.564	1.852	8.47	2.642	8.93
Seed 1405	0.564	1.826	7.34	2.677	7.22
Reco. mean	0.572	1.815	8.35	2.659	8.13
Reco. range	0.550–0.596	1.718–1.922	7.20–9.25	2.632–2.682	6.49–9.37

Table S47: Complete retrieval $\times$ construction design matrix explored on the released evaluator. Scores are corpus-level sacreBLEU-4 (0–100) on the canonical 641-item PHX-public set (REC=12.78). Of the 12 retrieval–construction cells, 6 were explored, 3 are not applicable (^a), and 3 were not explored (*, [†]); full reasons in `results/probe_multiplicity.json`. The headline probe (bold) was selected as the maximal-gap explored cell; its CI is post-selection descriptive. Two random-donor baselines (below the midrule) lie outside the 4×3 matrix.

Retriever	Construction	Explored	Score	Gap vs. REC
Source-text Jaccard	Pure donor copy	✓	23.02	+10.24
Source-text Jaccard	PT-composed	✓	18.86	+6.08
Source-text Jaccard	Oracle-gloss-comp.	—	—	N/A ^a
LaBSE	Pure donor copy	✓	19.2	+6.42
LaBSE	PT-composed	—	—	not explored*
LaBSE	Oracle-gloss-comp.	—	—	N/A ^a
multi-SBERT	Pure donor copy	✓	16.5	+3.72
multi-SBERT	PT-composed	—	—	not explored*
multi-SBERT	Oracle-gloss-comp.	—	—	N/A ^a
Oracle gloss	Pure donor copy	✓	11.4	−1.38
Oracle gloss	PT-composed	✓	11.8	−0.98
Oracle gloss	Oracle-gloss-comp.	—	—	not explored [†]
<i>Reference-free retrieval and random baselines (outside the 4×3 matrix)</i>				
Pose-signature NN (no text)	Pure donor copy	✓	0.45	−12.33
Random donor	Pure donor copy	✓	0.90	−11.88
Random donor	PT-composed	✓	0.77	−12.01

^aOracle-gloss-composed is definitionally tied to the oracle-gloss retriever, so it cannot be applied to a text-based retriever without changing the retriever. *Not explored; deemed low priority after Jaccard pure-copy dominated. [†]Subsumed by Oracle-gloss $\times$ PT-composed. Of the 12 retrieval–construction cells, 6 were explored, 3 are not applicable, and 3 were not explored. The headline probe was selected as the maximal-gap cell, so its CI is post-selection descriptive.

Table S48: Retrieval-query $\times$ scoring-frame grid under the released evaluator (461-item subset; sacreBLEU-4; descriptive paired bootstrap CIs, 10,000 resamples). The response is positive only on the diagonal where the retrieval query coincides with the scoring reference. The two selectors pick the same donor for only 6.5% of queries; human-query donors still average Jaccard 0.30 to the original reference texts (canonical selector: 0.55; human-query selector against the human text: 0.33).

	REC (recorded poses)	PURE, query=original	PURE, query=human
Score vs. original refs	14.94	24.98	10.92
Gap vs. REC	—	+10.03 [8.41, 11.69]	−4.02 [−5.50, −2.62]
Score vs. human refs	7.35	8.30	11.64
Gap vs. REC	—	+0.95 [−0.14, 2.02]	+4.29 [3.09, 5.49]

Table S49: Sensitivity grid (seed 42; protocols as in the main text). The CTC block varies the CTC denominator at clip 1.0; the clipping block varies `max_norm` with the per-sentence CTC default; the released row is the shipped evaluator. Readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate

Knob	Setting	Dev BLEU-4	Test REC	PURE	Gap	Readout (EM%)
CTC denom. (clip 1.0)	sentences	12.05	13.11	10.32	−2.80	11.96 (2.4%)
	gloss tokens	10.79	10.42	9.01	−1.40	10.06 (2.0%)
	input frames	9.55	9.62	8.75	−0.88	9.18 (1.9%)
	raw sum	0.90	0.95	0.95	+0.01	0.67 (0.0%)
Clip ∞ ? (CTC sent)	1.0 (seven-correction)	12.05	13.11	10.32	−2.80	11.96 (2.4%)
	2.0	17.17	16.35	15.90	−0.45	23.65 (9.0%)
	3.0	17.13	16.78	21.82	+5.04	53.21 (31.8%)
	5.0	16.54	15.92	23.97	+8.05	82.69 (68.4%)
	none (seed 42)	16.32	15.08	26.27	+11.19	99.98 (99.9%)
	none (seed 43)	15.78	14.32	25.63	+11.31	95.25 (90.5%)
	none (seed 44)	15.74	15.96	26.27	+10.31	99.98 (99.9%)
candidate-upstream	none + raw sum	16.04	15.12	25.80	+10.68	97.60 (95.1%)
Released evaluator		13.38	12.78	23.02	+10.24	78.8 (70.7%)

Dev-split consistency panel (results/dev_probe_eval.json).

The dev-split TN-PURE-v1 donor registry was constructed with the identical algorithm, frozen, and never used for probe construction or exploration; the released evaluator scores REC 13.38 / PURE 21.47 / gap +8.10 on it. The sensitivity checkpoints evaluated under the same frozen registry:

Tables relocated from the main text.

The tables below were relocated from the main text for length (main-text pointers: Tables S49–S51).

Table S50: Summary of the three primary descriptive comparisons over the full constructible family. Family beam-3-decoded training runs (range over 69 non-degenerate; 5 degenerate excluded); (ii) 38 constructible ch with uniform full-pool readout (released listed separately); (iii) gate-eligible non-distillation training runs. family composition in SI Sup. D. All canonical families share the clip=1.0 inference; the clipping sensitivity (outside this panel) extends the constructible ranges to readout 99.98 BLEU / 99.9% EM and gap +11.19.

Comparison	Family <i>n</i>	Released evaluator	Family range	Released – fa
(i) Gap–competence	74	+10.24 at dev 13.38	[−2.80, −0.21] (69 non-deg.)	
(ii) Train–pool read- out BLEU	38	78.8 (70.7% EM)	6.6–12.08 (1.3–2.5% EM)	
(iii) Competence gate (dev)	60	13.38	6.5–12.62 (seven-correction max 12.62; 1/8 pass)	

Table S51: Reproducibility accounting per headline claim. Layers: L1 recompute from committed decodes; L2 analysis re-execution; L3 re-decode from checkpoint weights; L4 from-scratch retraining. “Licensed” = verifiable end-to-end by a third party holding the PHOENIX-2014T and SLRTP2025 research licences.

Claim	Inputs beyond Layer	artifact	Licensed end-to-end	Compute / failure condition
REC 12.78, PURE 23.02, gap +10.24	none (per-item decodes committed)	L1	yes, licence-free	~10 CPU-min; fails if donor-registry SHA-256 $\neq$ 9170a530...
Faithful (Table 2)	family checkpoints (external mirror) or retrain	L3/L4	yes licences	with L3 $\approx$ 1 GPU-h/ckpt; L4 $\approx$ 6 GPU-h/seed; fails if decodes leave seed-dispersion range
Readout EM	78.8 / 70.7% released ckpt + train poses	L3	yes licences	with $\approx$ 1 GPU-h; fails if EM deviates
Readout-gap correlations (§4.2)	per-checkpoint readouts + gaps	L2	yes, licence-free	minutes; fails if ρ leaves CI
Reference-frame contrast 9.08	Czehmann CSV (CC BY-NC-SA, in artifact)	L1/L2	yes, licence-free	minutes; fails if matched subset $\neq$ 461
Step-matched / clip sensitivity (§??)	CTC / retraining	L4	yes licences	with $\approx$ 6 GPU-h/run; fails if the unclipped run’s readout or gap leaves the signature regime, or if CTC variants flip any conclusion

Table S52: Step-matched evaluation of fresh seven-correction runs at the released run’s best (1,820) and final (2,828) optimizer steps. Dev BLEU-4 is the training-time validation at the matched step; REC/PURE/gap use the canonical 641-query protocol; readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate. Released evaluator at its best step: dev 13.38, readout 78.8 (70.7%), gap +10.24.

Seed	Step	Dev BLEU-4	Test REC	PURE	Gap	Readout (EM%)
42	1,820	11.05	11.88	9.87	−2.00	10.85 (2.2%)
43	1,820	11.27	11.28	9.82	−1.46	11.23 (2.3%)
44	1,820	11.06	12.37	9.75	−2.63	11.51 (2.4%)
42	2,828	11.60	12.56	10.56	−2.00	11.81 (2.4%)
43	2,828	11.23	11.46	9.69	−1.77	11.32 (2.3%)
44	2,828	11.78	12.46	9.96	−2.49	11.61 (2.4%)

Table S53: Frozen-dev-probe evaluation (515 queries; beam-3; registry SHA-256 prefix 2d4780a61ac3). All rows are best checkpoints. Test-split gaps for the same checkpoints are in Table S49/main Table ??.

Evaluator	Dev REC	Dev PURE	Dev gap
ctc_clip_sens/clipinf_seed42	16.32	24.39	+8.07
ctc_clip_sens/clipinf_seed43	15.78	23.78	+8.00
ctc_clip_sens/clipinf_seed44	15.74	24.39	+8.64
ctc_clip_sens/candidate-upstream_seed42	16.04	23.91	+7.87
ctc_clip_sens/clip5p0_seed42	16.54	22.50	+5.96
ctc_clip_sens/clip3p0_seed42	17.13	21.27	+4.14
ctc_clip_sens/clip2p0_seed42	17.17	15.39	−1.79
seven-correction seed 42 (clip 1.0)	12.05	10.70	−1.35
Released (reference)	13.38	21.47	+8.10

Table S54: Implementation ladder from the original defective protocol to the seven-correction re-implementation and the unclipped resolution. All variants share the released architecture, data, and vocabularies; they differ in loss composition, normalisation, validation cadence, selection decode, stopping rule, and (final row) gradient clipping. Dev BLEU-4 is the uniform beam-3 full-pool value; gap is the canonical PURE-REC sacreBLEU-4 difference on the 641-item test set; readout is full-training-pool free-decode BLEU-4 (released: 78.8).

Protocol remaining)	(defects	Selection / validation	Dev BLEU-4	Readout	Gap range (neg. frac.)
Translation-only epoch / NLL (1–4, 5–7)	/	NLL / epoch	6.5–9.4	7.2–9.3	[−1.85, −0.80] (14/14)
validation-freq-misread (2 corrected)		NLL / epoch	8.3–9.8	—	[−1.70, −0.52] (4/4)
Translation-only / step-val / BLEU (1, 4, 5–7)		greedy / 14 steps	7.6–10.5	—	[−1.37, −0.36] (10/10)
Joint-loss / step-val / greedy-BLEU (4–7)		greedy / 14 steps	6.3–10.2	—	[−1.59, −0.48] (8/8)
Joint-loss / step-val / beam-3 (5–7)		beam-3 / 14 steps	8.7–10.5	—	[−1.38, −0.37] (8/8)
Seven-correction (7 corrected; clip 1.0; seeds 42–49)		beam-3 / 14 steps	11.59–12.62	11.45–12.08	[−2.80, −1.37] (8/8)
Seven-correction, unclipped (8th corrected; seed 42)	cor-	beam-3 / 14 steps	16.32	99.98	+11.19
Released evaluator	—		13.38	78.8	+10.24

Table S55: CTC-term normalisation sensitivity (seed 42, clip 1.0, otherwise the seven-correction protocol). REC/PURE/gap use the canonical 641-query protocol; readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate. None of the four denominators matches the released log’s step-14 Recognition Loss of 17.28 (§3.4); the raw-sum degeneracy holds only at clip 1.0 (see the candidate-upstream run, Table ??).

CTC denominator	Dev BLEU-4	Test REC	PURE	Gap	Readout (EM%)
sentences (default)	12.05	13.11	10.32	−2.80	11.96 (2.4%)
gloss tokens	10.79	10.42	9.01	−1.40	10.06 (2.0%)
input frames	9.55	9.62	8.75	−0.88	9.18 (1.9%)
raw sum	0.90	0.95	0.95	+0.01	0.67 (0.0%)

Multi-seed clip-ladder replicates.

Seeds 43–44 replicate the intermediate clip thresholds (`results/faithful_steps_eval.json`, `results/dev_probe_eval.json`). The seed-43/44 runs terminated at the 1,200-validation hard safety cap rather than the LR-floor trigger: by termination the LR was frozen at the scheduler’s numerical floor ($4.36\text{e-}8$, the same terminal state as the published family) and the best checkpoint was unchanged over the final $\gtrsim 900$ validations, so the cap does not affect best-checkpoint selection. Across the three seeds per threshold, readout, test gap, and dev-probe gap are each monotone in the threshold; for the test gap and the dev-probe gap every adjacent threshold pair is separated by more than the between-seed range at either threshold, while in readout the clip-5.0 and unclipped ranges overlap (82.69–99.80 vs. 95.25–99.98) because the readout saturates near the ceiling.

Table S56: Multi-seed clip ladder (three seeds per threshold; seed-42 rows correspond to main Table ??). Dev BLEU-4 is the training-time best validation; REC/PURE/gap use the canonical 641-query protocol; readout is full-training-pool free-decode BLEU-4 with exact-match rate; dev-probe gap uses the never-explored dev split registry. Mean $\pm$ SD over the three seeds.

Threshold	Seeds	Test gap	Readout (EM%)	Dev-probe gap	Dev BLEU-4	Test REC
1.0	42–49 (8)	$[-2.80, -1.37]$	11.45–12.08 (2.3–2.5)	s42: -1.35	11.59–12.62	11.60–13.11
2.0	42–44	-1.11 ± 0.63	23.2 ± 0.5 (8.5–9.0)	-1.81 ± 0.26	17.10 ± 0.44	16.65 ± 0.58
3.0	42–44	$+5.39 \pm 0.69$	56.5 ± 4.3 (31.8–40.3)	$+3.90 \pm 0.22$	17.34 ± 0.48	16.77 ± 0.24
5.0	42–44	$+9.28 \pm 1.07$	92.5 ± 8.8 (68.4–99.0)	$+6.85 \pm 0.76$	16.77 ± 0.29	16.06 ± 0.16
none	42–44	$+10.94 \pm 0.55$	98.4 ± 2.7 (90.5–99.9)	$+8.24 \pm 0.35$	15.95 ± 0.32	15.12 ± 0.43
candidate-upstream	42	+10.68	97.60 (95.1)	+7.87	16.04	15.12
Released	—	+10.24	78.8 (70.7)	+8.10	13.38	12.78

24 Update-Level Gradient Diagnostics

Question answered: the clip ladder (Sup. U) establishes that the two signatures are clipping-regime properties, but raw pre-clip gradient norms do not show how the clip acts on the *realised* optimiser behaviour, because gradient clipping interacts with the batch-normalised loss scale, the CTC term, Adam’s first and second moments, and weight decay. This appendix measures that interaction directly.

Design. `src/training/train_grad_diag.py` trains instrumented replicas of the seven-correction protocol (seed 42, otherwise identical) while logging, at every optimiser step: the pre-clip gradient norm, the post-clip norm, the realised parameter-update norm $\|\theta_t - \theta_{t-1}\|_2$, the cosine between the update vector and the raw pre-clip gradient, and—every 14th step, on a shared retained computation graph—the translation-gradient and CTC-gradient component norms. Three runs:

1. *Dual lockstep run* to step 2,828 (the released run’s final step): a clip-1.0 arm and an unclipped arm trained simultaneously from the identical seed-42 init over the

identical batch order with identical dropout masks (the CUDA RNG state is forked per step), so matched-step update norms and update-direction cosines between the two regimes are directly comparable. The clipped arm consumes the same RNG stream as a standalone `train_faithful.py` run and replicates its published gradient norms step-for-step (377.53, 568.21, 145.46, ...), so it doubles as a determinism check.

2. *Single clipped run* to its natural termination (full-horizon clipped-step proportion).
3. *Single unclipped run* with an `--lr` override for the update-norm-matched control (Adam steps scale linearly in the learning rate).

Outputs: `results/grad_diag_dual_seed42.json`, `results/grad_diag_clipfull_seed42.json`, `results/grad_diag_unclipfull_seed42.json` (+ control), with per-step records and summary statistics; run directories under `checkpoints/grad_diag/`.

Results (dual lockstep, 2,828 steps). The clipped arm’s pre-clip gradient norm exceeds the 1.0 threshold at *every* step of the matched window (2,828/2,828; median 37.6, p10–p90 26.3–42.7), so the post-clip norm is pinned at 1.0 throughout; the unclipped arm’s pre-clip norms are smaller in median (20.3) because its parameters occupy a different loss region. The realised update norms tell the mechanistic story: at step 1 both arms take updates of essentially identical norm (2.415 vs. 2.417) despite the $377\times$ gradient reduction; over the first 200 steps the unclipped/clipped update-norm ratio averages 0.89, and over the full window its median is 2.12—the clipped run’s updates are *not* proportionally smaller, and end up the smaller ones, a comparison that conflates the two regimes’ diverging LR schedules (the unclipped arm’s higher dev BLEU delays plateau decays, so its LR stays larger; we report the ratio with this caveat). The update directions rotate apart: the cosine between the two arms’ matched-step updates is 0.43–0.55 over the first ten steps (mean 0.45 over the first 100) and decays to a window median of 0.06; within each arm, the cosine between the update and the raw pre-clip gradient is near zero (clipped arm mean -0.005 ; unclipped -0.24), i.e. Adam’s per-coordinate normalisation already rotates the update far from the gradient direction in both regimes. Gradient-component norms (every 14th step): the translation and CTC components are of comparable magnitude in both arms (clipped-arm medians 30.7 and 21.9; unclipped 16.6 and 14.4), with the CTC component dominating the unclipped arm’s earliest steps (38.3 vs. 12.1 at step 14)—the large pre-clip norms are a joint property of the two terms, not a CTC artefact. The full-horizon single runs corroborate the proportion: the clipped run clips at every one of its 8,120 steps to natural termination (pre-clip norm median 46.4), and the unclipped run terminates at 7,840 steps with best dev BLEU-4 16.32 at step 2,352—both replicating the published family’s terminal behaviour.

Interpretation. The first Adam update is scale-invariant under uniform gradient rescaling, so the $377\times$ gradient reduction at step 1 produces updates of essentially identical norm; the clip acts instead through the weight-decay term (which the small clipped gradient no longer dominates in the effective gradient), through the second-moment history, and through the trajectory-level rotation of update directions. Over the full window the clipped run’s updates are the smaller ones, a comparison that conflates the regimes’ diverging LR schedules; the update-norm-matched control separates update magnitude from regime: retrained unclipped at $\text{lr} = 5.5 \times 10^{-4}$

($= 10^{-3} \times 0.066 / 0.120$, so its realised update norms match the clipped run’s median), the control still enters the memorisation regime (readout 84.91 BLEU / 75.1% EM, probe gap +9.51, dev 15.68 at its best checkpoint; canonical-protocol evaluation, row `ctc_clip_sens/lrmatched5p5e4` of `results/faithful_steps_eval.json`). The claim the diagnostics support is deliberately operational: training dynamics in our reconstruction are highly sensitive to the clipping setting, and the effect operates through the optimiser’s interaction with the clipped gradient scale—not through a proportional shrinkage of the parameter update, and not through update magnitude alone. All diagnostics are single-seed (42) trajectory-level measurements and do not settle the released run’s own training mechanism (main §??).

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